The Deadly Race

When Graceful Recovery Makes Coercion-Resistant Custody Worse Than Useless

A formal threat model for physical-coercion ("wrench") attacks, and a construction that clears it

Great Wall coercion-resistance paper series

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Abstract

This paper contributes a *design bar* for coercion-resistant custody and a *construction* that meets it, motivated by a sharp negative result: custody designs that lack such a bar can be *worse than useless*, actively endangering the holder they claim to protect. Physical-coercion ("wrench") attacks defeat custody schemes that key management and cryptography treat as out of scope. We give a *formal model* of coercion against a secret-holder and prove that a large class of "graceful" recovery designs — those that leave a device seizure holding a *feasible-but-slower* path to the secret — are strictly worse than useless: they manufacture a material incentive to *assassinate* the victim. We name this the **Deadly Race**, derive it from a single impossibility (one cannot prove one has forgotten), and distill a design bar — the **No-Gray-Area Criterion** — that any coercion-resistant scheme must clear. We then present a construction — the **Time-Gated Perceptual Oracle** (TGPO) — that clears the bar for the Bitcoin self-custody setting while keeping custody *individual*, and we locate a shipping alternative — timelock-rescue vaults — precisely: it clears the same bar by the *other* available route, at the cost of a trusted third party. The framework fixes a bar; it does not claim TGPO is the unique construction that clears it.

Keywords: Bitcoin; self-custody; wrench (physical-coercion) attacks; coercion resistance; Kerckhoffs's principle; Deadly Race; No-Gray-Area Criterion; Tacit Knowledge-Based Authentication; Time-Gated Perceptual Oracle; memorization; hostage token; shared custody; geographic custody.

1 Introduction

Coercion is the unmodeled attack. Standard threat models — Dolev–Yao on the wire [1], IND-CCA for encryption, forward secrecy for sessions — all assume the adversary cannot compel the *holder* of a secret. For bearer assets that assumption is empirically false: the physical attack registry maintained by Lopp [2], and the systematic study of the phenomenon in the financial-technologies literature [3], document a growing series of incidents in which a Bitcoin holder is physically threatened with a "$5 wrench" until they surrender their keys.

Cryptographic hardness does not help here. A wrench attacker does not break the cipher; they compel the person who can operate it. Worse, the standard defense offered to holders — *obscurity* ("never tell anyone you own Bitcoin") — is not a security mechanism at all in the sense of Kerckhoffs [4]: it is a bet that the attacker lacks knowledge the defender has, and it degrades precisely as adoption spreads that knowledge. This "keep a low profile, never disclose" counsel is nonetheless the community's entrenched consensus, actively promoted by prominent voices (catalogued in the accompanying economic analysis [5]); and it carries a measurable price — that analysis finds the severity-adjusted actuarial cost of individual Bitcoin self-custody already *exceeds* the cost priced into insurance for comparable portable valuables such as jewellery.

This paper models coercion directly. Our contributions are:

1. a **formal coercion model** (§2) with an explicit adversary, a Kerckhoffs premise, and an engineering "feasibility line" that turns asymptotic security into concrete parameter bounds;
2. the **Deadly-Race impossibility chain** (§3): a classical impossibility (you cannot prove you forgot) that escalates, step by step, into an incentive to murder the victim;
3. the **No-Gray-Area Criterion** (§3), the design bar that a coercion-resistant scheme must clear, together with a *reachability* dichotomy that classifies every clearance route;
4. a **construction** (§4), the TGPO orbit, that meets the bar with individual custody, including a proof that the master secret meets the floor from every seizable state bar a bounded residual window; and
5. a **taxonomy** (§5) that classifies existing designs by *which* clearance route they take — illustrated on shipping time-lock-rescue vaults.

We call the central results the **Deadly-Race Lemma** and the **No-Gray-Area Criterion**, after their first statement in the Great Wall design record [6].

2 Model

Notation

The symbols below recur throughout; each is defined in full where it is introduced, and all are gathered here for reference.

Principals, assets, economics	
$\mathcal{V}$, $\mathcal{A}$	holder (victim) and coercer (attacker)
K	master secret gating the asset
W	value of the asset to $\mathcal{V}$
c	expected disutility to $\mathcal{A}$ of killing $\mathcal{V}$
π	$\mathcal{V}$'s *pivotality* — win-probability $\mathcal{A}$ gains by removing $\mathcal{V}$
Adversary, game, safety	
WDY	the Wrench–Dolev–Yao adversary, fixed by clauses D0–D8 (Assumption 2)
p_A	probability $\mathcal{A}$ reaches K after the encounter; *gray* iff $0 < p_A < 1$
t	coercion-time budget (how long $\mathcal{A}$ can sustain coercion)
G	protocol time-gate (enforced latency to complete the derivation)
Corr	residual-window event: the points $\mathcal{V}$ has surrendered so far are correct (§4.7)
Esc	residual-window event: the ≈1–2 min shaved by brute-forcing the last points decides escape from the scene (§4.7)
Construction (TGPO orbit §4)	
σ	public, precomputation-resistant timechain salt (Namtso)
H, H^*	unspecified cheap hash; unspecified memory- and depth-hard (Argon2-class) hash
o_i	orbit state at stage $i \in \{0, \ldots, N\}$; $o_i = \mathsf{H}^*(u_{i-1})$ (with $o_0 = \mathsf{H}(\sigma)$); $N{+}1$ stages total
N	number of *deep* stages — those requiring at least one memory- and depth-hard hash (H^*) to derive
Sh_i	Shamir polynomial at stage i; r_i of its points reconstruct it
u_i	$\mathsf{H}(o_i \Vert Sh_i)$: commitment of the wiped values o_i and Sh_i
K_i	$\mathsf{H}(o_i \Vert Sh_i)$: master secret correspondent to stage i.
$p_{i,j}$, $\theta_{i,j}$	point j of stage i; parameters for fractal j of stage i
r_i	Shamir threshold at stage i ($r_i \geq 2$)
64^*	2^{64} memory- and depth-hard evaluations (infeasible)
floor	*prohibitive* cost: $\geq 64^*$ hard *or* $\geq 2^{96}$ cheap (Stipulation 1)

2.1 Principals and capabilities

Definition 1 (Principals)**.** The **holder** $\mathcal{V}$ possesses (in memory and on devices) whatever is needed to reach a master secret K that gates an asset. The **coercer** $\mathcal{A}$ seeks K. Optional **heirs** or **delegates** may hold partial capability on $\mathcal{V}$'s behalf.

Assumption 1 (Kerckhoffs, premise 0)**.** $\mathcal{A}$ knows the protocol in full: all algorithms, all source, $\mathcal{V}$'s setup topology, and every in-scope parameter. Only the protocol *secrets* are hidden. Every result below is proved *against* such an $\mathcal{A}$; security by obscurity is assumed to contribute nothing.

Assumption 2 (The Wrench–Dolev–Yao (WDY) adversary)**.** The **Wrench–Dolev–Yao (WDY)** adversary inherits the Dolev–Yao [1] intruder *on the wire* — it sees all transmitted plaintext and ciphertext — but departs from it in two ways that matter here. It is *computational*, not symbolic: Dolev–Yao's perfect-cryptography idealization is replaced by the feasibility line (Stipulation 1), so "cannot break the cipher" becomes "cannot exceed its work budget". And it is extended *past the wire* with the physical capabilities Dolev–Yao omits — device seizure and a bounded coercion oracle over the holder: the *wrench* the name records. WDY is fixed by:

- **D0. Kerckhoffs.** $\mathcal{A}$ knows the protocol, $\mathcal{V}$'s setup topology, and every in-scope parameter (Assumption 1).
- **D1. Bounded compute.** $\mathcal{A}$'s computation is bounded by the feasibility line (Stipulation 1 below): 2^{32} cheap evaluations feasible, plain 2^{64} cheap at the border (the softest number), and 2^{64} memory-hard (64^*) and 2^{96} cheap both infeasible. This is what replaces "unbounded": the wall-clock custody window bounds $\mathcal{A}$ crudely, the feasibility line *sharply*.
- **D2. Frequency ceiling.** There is a ceiling on how much faster $\mathcal{A}$'s hardware evaluates a computation than $\mathcal{V}$'s. So an entropy-preserving hash with enough *non-parallelizable* steps (H^*) lets $\mathcal{V}$'s setup impose an *arbitrarily long* latency on $\mathcal{A}$ per evaluation — at the price of a proportionally longer latency on $\mathcal{V}$. This is what makes the time-gate meaningful: $\mathcal{A}$ can neither parallelize nor out-clock its way past a sequential memory-hard chain.
- **D3. Complete subjugation.** This is an information-security model, not a study of physical violence versus physical defense; we do not model $\mathcal{A}$'s coercive power against $\mathcal{V}$'s resistance. For simplicity, and to make the argument *a fortiori*, we assume that from the onset of the attack until $\mathcal{A}$ releases $\mathcal{V}$, $\mathcal{A}$ meets *no resistance whatsoever* and enjoys $\mathcal{V}$'s *full cooperation.* The sole qualification — *feigned* cooperation in the residual window — is, as we argue there (§4.7), of residual importance.
- **D4. Co-located device seizure.** All of $\mathcal{V}$'s devices *at the attack site* are immediately seized — vault blobs, time-lock blobs, commitment scripts, review schedules. A single WDY encounter has one *site*; devices and principals at *other* sites lie beyond its reach within the bounded window — the opening a *geographic* defense exploits (§3.6).
- **D5. Explicit-secret seizure.** All of $\mathcal{V}$'s *explicit* (non-tacit) authenticating secrets are immediately seized.
- **D6. No tacit seizure.** $\mathcal{V}$'s *tacit* entropy-carrying secrets — the points — are captured by seizure *if and only if* they are *available* in transmissible form at the moment of attack (materialized, and not wiped before $\mathcal{V}$ is subdued) — that is, exactly when the tacitness fails to be effective. When it *is* effective they are not: by D3 this is a genuine *incapacity to transmit*, not any unwillingness — even a fully cooperating $\mathcal{V}$ cannot hand them over, because they are recognizable but non-transmissible. This and D7 are the operational restatement, in the adversary model, of the tacit-knowledge premise (Assumption 3).
- **D7. No perceptual-parameter seizure.** As with the tacit entropy-carrying secrets (D6), $\mathcal{V}$'s perceptual-oracle parameters — the deep fractals — are seized *iff* they are available at the moment of attack; when the tacitness is effective they are not. As in D6, and by D3, this is incapacity under full cooperation, not withholding: the deep fractal parameters are regenerated only by walking the memory-hard chain, so a willing $\mathcal{V}$ has nothing to dictate (Assumption 3).

D8. No remote-delegate coercion. A remote delegate — a party holding rescue capability on $\mathcal{V}$'s behalf beyond the attack site (D4) — is past $\mathcal{A}$'s *remote* coercive reach: from the encounter with $\mathcal{V}$, $\mathcal{A}$ cannot compel it to surrender or act on its state. Without this a delegated defense is vacuous. Real delegates span a spectrum of coercion-compliance; collapsing it to a clean "cannot be reached" is a deliberate simplification — the kind that makes the *a fortiori* argument possible, at the cost of reality's shades.

The *only* channel to the tacit material withheld by D6–D7 is a **bounded coercion oracle**: $\mathcal{A}$ extracts it through the cooperating $\mathcal{V}$ (D3) over a bounded duration t — the adversary's **coercion-time budget**, the maximum wall-clock time for which it can sustain coercion, calibrated against the incident durations in the registry [2]. Cooperation is genuine throughout by D3; the one exception, *feigned* cooperation, is confined to the residual window (§4.7).

Remark 1 (The two non-seizure clauses coincide for design). For protocol-design purposes the tacitness of the entropy-carrying secrets (D6) and that of the perceptual-oracle parameters (D7) are logically equivalent. If the parameters were transmissible, a wrench attack would seize them — and, available within that *same* encounter, they would then serve to seize the entropy-carrying points themselves; so non-tacit parameters force non-tacit points. Conversely, if the points were somehow transmissible without the parameters ever being available, then the tacitness of the parameters is irrelevant, the points having already fallen. Either way the binary question "does a single seizure reach K?" turns on the two clauses together, and the model may treat them as one.

Remark 2 (Where the distinction still matters). The distinction can nonetheless matter for a *quantitative* refinement — a statistical distribution of the wall-clock needed to coerce either (up to sufficient number of most-significant bits) given $\mathcal{A}$'s coercion time budget t (Definition 4). Still, a secret that is *theoretically* transmissible, but only with pen and paper, an adversarial fork of the application, and long hours to days of focused mental effort under perfect $\mathcal{A}$–$\mathcal{V}$ cooperation, is essentially as good as ideal. Until empirical studies of TKBA (Assumption 3) are done, the actual relevance of that nuance is purely speculative.

2.2 The feasibility line

Asymptotic "hard" and "easy" are useless against a fixed, resourced attacker; we replace them with two engineering stipulations that fix what *feasible* means and turn the lemmas into concrete bounds.

Stipulation 1 (Feasibility line)**.** Let a *cheap* evaluation be one call to a non-memory-hard hash H, and a *hard* evaluation one call to an hours-scale Argon2-class function H^* whose cost is a large number of *non-parallelizable* steps — *memory- and depth*-hard,[1] so $\mathcal{A}$'s bounded speed advantage (D2) cannot amortize it [8].

S1 (feasible). 2^{32} cheap evaluations are within $\mathcal{A}$'s budget.

S2 (infeasible). 2^{64} *hard* evaluations — written $\mathbf{64}^*$ — are beyond it.

S3 (infeasible). 2^{96} cheap evaluations are beyond it.

The star denotes this memory- and depth-hardness and attaches to *any* bit-count — $32^*, 64^*, 96^*, \ldots$ — each meaning that many H^* evaluations, not just 64^*.

The floor. We call a work cost *prohibitive* — equivalently, say it *meets the floor* — when it satisfies S2 *or* S3: at least 64^* memory-hard evaluations, *or* at least 2^{96} cheap ones. The two bounds are independent, and we deliberately fix *no* ordering between them. For realistic memory-

[1] *Depth-hard* is meant more sharply than *sequential* or *non-parallelizable* alone: the cost is a non-parallelizable computation with a *deliberately large number of steps* — a long sequential *depth* — the parallelism-resistance axis of *depth-robust* memory-hard functions [7] — that $\mathcal{A}$ can neither parallelize away nor out-clock past (D2). We also weighed *memory- and sequentiality-hard* (accurate but clumsy) and *memory- and iteration-count-hard* (which names the step *count* rather than its non-parallelizable structure); *depth* is the single noun that carries both the seriality and the magnitude, and it parallels *memory*.

and depth-hardness the per-evaluation $*$ premium is well under 2^{32}, so $64^* < 2^{96}$ and the cheap bound is the binding one; but $*$ is a user-tunable lever, and an aggressive enough setting pushes that premium past $2^{32} \approx 4.29 \times 10^9$ — above 32 extra bits — making $64^* \geq 2^{96}$. Every "floor" claim below names *this* threshold, not a fixed number, and holds whichever bound binds.
We keep the star rather than fold it into an equivalent bit-count because memory- and depth-hardness has *three* distinct consequences, only the first of which a bit-count could absorb: it **(1)** inflates the brute-force cost — this alone *could* be traded for a stipulated increase in bits; **(2)** imposes a wall-clock *derivation time* on each evaluation, which the bounded coercion window t (D3) cannot outrun; and **(3)** rules out the quantum (Grover) speed-up (§6). A pure bit-count captures only (1); the star carries all three.

Two currencies span the gap between them. *Entropy* buys $32 \to 64$: a second independent 32-bit stage. But plain 2^{64} *cheap* work still sits at the border of a resourced attacker's reach (hours), so the $64 \to 64^*$ step must be bought by *memory-hardness* — which is why the gating function has to be Argon2-class. We refer to plain cheap 2^{64} throughout as the *softest number*: it must be named explicitly wherever it is invoked, never silently treated as infeasible.

2.3 The tacit-knowledge premise

The construction of §4 rests the secret on *tacit* knowledge. We state the premise abstractly here; §4 instantiates it.

Assumption 3 (Tacit knowledge, TKBA)**.** *The secret is a set of fractal locations that are recognizable but not transmissible.*

- **TKBA$_1$.** A point is non-transmissible without the fractal parameters that render it; $\mathcal{V}$ cannot dictate it in the abstract.
- **TKBA$_2$.** The *deep* fractal parameters are not a hand-overable artifact; producing them requires walking a memory-hard chain, before which later locations do not exist.

Remark 3 (TKBA restates the tacit-seizure clauses)*.* Assumption 3 is not an independent hypothesis: it is clauses D6–D7 lifted out of the adversary model and stated as a premise on the secret. The clauses say $\mathcal{A}$ cannot seize the tacit points or the perceptual parameters even under full cooperation (D3); TKBA says the same thing about the secret itself — recognizable but not transmissible — with no adversary mentioned. They are two faces of one assumption (exactly as D0 and Assumption 1 are two faces of Kerckhoffs), and by Remark 1 the two clauses are themselves one. We keep both forms because each is the natural one in its place — the premise where the construction is motivated (§4), the adversary clause where seizure is reasoned about.

3 The Deadly-Race impossibility chain

"Nothing was your own except the few cubic centimetres inside your skull."
— George Orwell, *Nineteen Eighty-Four*

This section states the paper's core negative result as an *escalating chain*: the *Non-Negative-Proof Principle*, an *a fortiori* assumption on the attacker's knowledge (justified by an information-theoretic impossibility, §3.2), its decision-theoretic consequence for a rational coercer (§3.3), and the design bar the two together force (§3.4–§3.5). We are deliberately explicit about which links are *impossibilities* and which are *incentive* claims: the Deadly-Race Lemma is **not** a cryptographic hardness statement, and reading it as one would be exactly the overclaim §8 warns against.

3.1 Setup and definitions

Fix the model of §2. Write K for the spend authority (the Bitcoin key) protecting an asset worth W to the holder $\mathcal{V}$, and c for the expected cost to $\mathcal{A}$ of killing $\mathcal{V}$ — the legal, moral, and

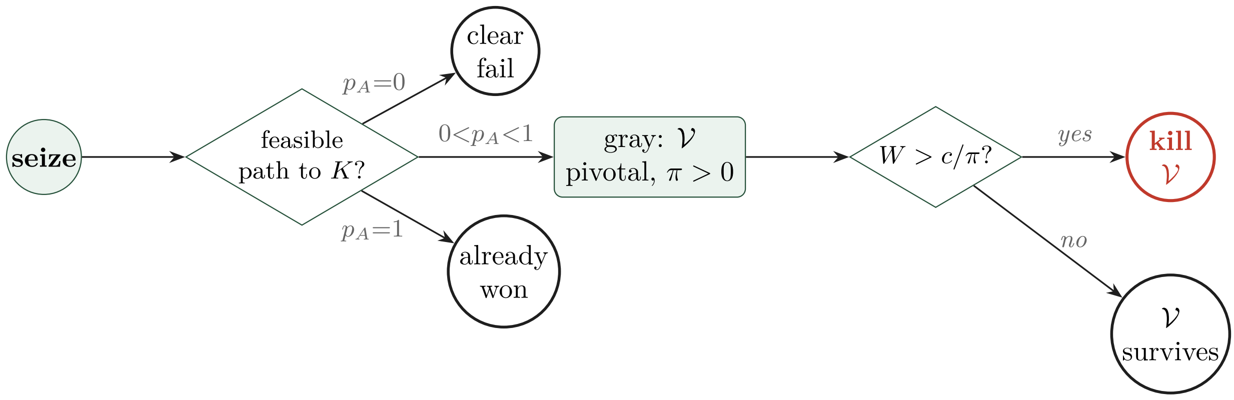

Figure 1: **The Deadly Race decision.** After a WDY seizure (Assumption 2), the outcome is fixed by p_A, the probability $\mathcal{A}$ reaches K. At the *clear* endpoints ($p_A \in \{0, 1\}$) no race exists. In the *gray* interval $0 < p_A < 1$ the holder $\mathcal{V}$ is *pivotal* ($\pi > 0$, Theorem 1): removing $\mathcal{V}$ raises $\mathcal{A}$'s win-probability, so once the asset clears $W > c/\pi$ a rational $\mathcal{A}$ has a positive incentive to *assassinate* (Lemma 1). Clearing the No-Gray-Area Criterion (Criterion 1) means removing the gray branch entirely.

operational disutility of a homicide. Bitcoin makes the prize *rivalrous* and *bearer*: the *first* party to broadcast a valid spend takes W; there is exactly one prize and no appeal.

Definition 2 (Seizable state, feasible path, racers)**.**

- **Seizable state Σ.** Everything $\mathcal{A}$ obtains from the encounter — all seized device state, plus whatever $\mathcal{A}$ extracts from $\mathcal{V}$ during the bounded coercion window.
- **Feasible path.** A *recovery path* from a state is an algorithm that outputs K within the feasibility line (Stipulation 1). A state *feasibly reaches* K if such a path exists; otherwise reaching K from it is *infeasible*.
- **Post-release racers.** If $\mathcal{A}$ releases $\mathcal{V}$, two parties may pursue K: $\mathcal{A}$ from Σ, and $\mathcal{V}$ from memory / tacit recall. Let $p_A \in [0, 1]$ be the probability that $\mathcal{A}$ broadcasts a valid spend *before* $\mathcal{V}$ does, given the states each holds after release. Because the prize is rivalrous, $\mathcal{V}$'s countermove — spending first, to safety — is $\mathcal{A}$'s only loss condition once $\mathcal{A}$ holds *any* feasible path.

Definition 3 (Gray vs. clear outcomes)**.** A coercion outcome is **gray** if afterward $0 < p_A < 1$: $\mathcal{A}$ has a feasible path ($p_A > 0$) yet $\mathcal{V}$, alive and free, can still win the race ($p_A < 1$). The **clear** outcomes are the endpoints — $p_A = 1$ ($\mathcal{A}$ certainly wins) and $p_A = 0$ ($\mathcal{A}$ has no feasible path at all). Figure 1 traces the decision this induces.

Definition 4 (Priced and sharpened adversaries)**.** The **priced** WDY adversary, written WDY$[W, c]$, is a WDY adversary (Assumption 2) additionally endowed with the economic parameters (W, c) and modeled as a *cold expected-utility maximiser*: it chooses among release, continued coercion, and assassination so as to maximize its assessed expected payoff over the deadly-race probabilities (the win probability p_A, and the correctness/escape events introduced in §4.7). Plain WDY carries *no* such payoff: it is the adversary of the general results (NNPP, the No-Gray-Area Criterion, the floor), which must hold whatever (W, c) turn out to be. WDY$[W, c]$ is invoked only where a *quantitative* incentive is actually computed — the Deadly-Race Lemma's condition (§3.3) and the residual-window analysis (§4.7).

A third, optional parameter sharpens the feasibility line to a single threshold. WDY$[W, c, b]$ replaces the two stipulations of Stipulation 1 by one *clear-cut* bound b — total work below b feasible, at or above b infeasible — in cost-homogeneous units, with memory-hardness folded into the per-evaluation cost. This idealization is legitimate here because the protocol quantizes entropy in 32-bit chunks, so feasibility is a step function on the lattice $\{32, 64, 96, \ldots\}$ and its hard counterparts $\{32^*, 64^*, 96^*, \ldots\}$: a threshold at that granularity loses nothing, whereas pricing a single brute attempt would be finer than any distinction the construction can express. Conservatively $b = 64^*$ — work below 64^* feasible (the softest number handed to $\mathcal{A}$), 64^* itself infeasible. A fourth optional parameter t budgets the **coercion time** (D3's window): WDY$[W, c, b, t]$ fixes the maximum wall-clock time for which the adversary can sustain coercion. Like b, t is a capability bound pinned at its real, registry-calibrated value t_{reg}, not sent to ∞ — unbounded coercion would let $\mathcal{A}$ walk any time-gate to completion, the same category error as $b \to \infty$. The rungs are thus WDY (capabilities only), WDY$[W, c]$ (+ payoff), WDY$[W, c, b]$ (+ sharp feasibility), and WDY$[W, c, b, t]$ (+ coercion-time budget).

Remark 4 (Plain WDY as a worst case). Plain WDY is exactly the worst case of the priced family over the *payoff* coordinates, taken at the model's *fixed* capability:

$$\mathrm{WDY} \;=\; \mathrm{WDY}[\,W \to \infty,\;\; c \to 0,\;\; b = 64^*,\;\; t = t_{\text{reg}}\,].$$

The payoff coordinates are sent to their extreme — infinite stake, costless homicide: the maximally kill-prone attacker, which assassinates in *every* gray outcome (§3.3) — because the general results must hold whatever (W, c) are. The capability coordinates are **not** worst-cased: b stays pinned at the feasibility line 64^* and t at the registry window t_{reg}. Sending $b \to \infty$ would grant $\mathcal{A}$ a 64^* brute and collapse the floor (Theorem 3); sending $t \to \infty$ would let it walk any time-gate to completion and collapse the tacit route (§3.6). Either is a category error, not a stronger guarantee. The asymmetry has a tell at the residual window: there $p_A \to 1$ while the worst case wants $W \to \infty$, making $(1 - p_A)\,W$ an $\infty \cdot 0$ indeterminate form — which is precisely why that window is settled with *finite* (W, c) and the Δp_A increment (§4.7), not with the blunt limit.

3.2 The Non-Negative-Proof Principle

Assumption 4 (Non-Negative-Proof Principle, NNPP — *a fortiori* on Kerckhoffs)**.** Kerckhoffs (Assumption 1) grants $\mathcal{A}$ the *protocol* — and that $\mathcal{V}$ *may* hold brain-memory or an undisclosed backup short-cutting K — but *not* how many such backups $\mathcal{V}$ keeps, or where. NNPP covers exactly that gap: what $\mathcal{A}$ *cannot* know is whether any undisclosed backup *actually* exists. The two do not conflict — one is the public mechanism, the other $\mathcal{V}$'s private state — so NNPP *qualifies* Kerckhoffs, and we resolve the gap conservatively: **$\mathcal{A}$ always assumes an undisclosed backup exists**. Notice that something as small as an SD card, as inconspicuous as steganographic notes in a book, or as interior as the memorized login and password of an email account suffices to store a short-cutting state. Again, for the purposes of the *a fortiori* argument, we do not consider the attacker reading truthfulness off $\mathcal{V}$'s terrorized demeanor or tone.

Both halves point the *same* way — toward killing. Knowing the protocol (Kerckhoffs) and always presuming $\mathcal{V}$ a live racer (NNPP) each *raise* $\mathcal{A}$'s assessed deadly-race payoff $\pi\,W$, hence its proclivity to kill — precisely the outcome the method must survive. Each is therefore *a fortiori*: we endow $\mathcal{A}$ with the *most kill-prone* epistemics and demand safety even then. In one line:

> *if, even after maximizing the attacker's proclivity to enact the deadly race, killing still buys it nothing, the method is* (a fortiori) *deadly-race-safe.*

In practice NNPP reduces to a plain observation: **exhaustive search is so ineffective that $\mathcal{A}$ always assumes another backup exists, whatever the terrorized $\mathcal{V}$ pleads.** It is rational, not paranoid — non-possession is a *negative existential* ("nothing anywhere in $\mathcal{V}$'s reach reproduces K") that cannot be certified on either reading of "prove."

Remark 5 (Why even total search fails — the self-defeat). The search reading fails not merely because the covering is large but because the capability that would complete it is *off the domain of the problem.* To certify $\neg\exists$ witness, $\mathcal{A}$ must reach the last, irreducible site — the "few cubic centimetres" of the epigraph: whether $\mathcal{V}$ has *forgotten*, or never committed, K to memory. An $\mathcal{A}$ that can read that space, and cover all physical hiding-space besides, is *supra-totalitarian* — it exceeds the literary supremum of surveillance power, which expressly conceded the skull — and it is a *degenerate fixed point* of the model: it violates the bounded-adversary premise so completely that the contested-property premise collapses with it, for under an uncontested overlord there is no custody to resist. Defending against it is a *doomsday bet*: the branch in which the adversary exists is the branch in which the stakes that define the game are gone, so a rational actor assigns it *zero decision-weight* — not because its probability is negligible, but because its utility is undefined there. NNPP is therefore safe as a *modeling boundary*: the only world that falsifies it is one in which the question does not arise.

Corollary 1 (Cooperation cannot dispel the incentive)**.** *Let b be the number of resumable copies $\mathcal{V}$ holds, and under the coercion oracle (D3, Assumption 2) let $\mathcal{V}$ disclose some of them. Each disclosure* confirms *a specific copy existed, raising a known lower bound $\hat{b} \leq b$; none yields an upper bound $\hat{b} \geq b$, which would require the covering search of Remark 5. Disclosure is thus a* monotone lower-bound estimator *— "$\mathcal{V}$ has revealed everything" is precisely the direction NNPP forbids. Hence no degree of cooperation drives $\mathcal{A}$'s posterior on "≥ 1 undisclosed copy" to zero: in any design that leaves a released $\mathcal{V}$ some resumable path, the assassination incentive of the Deadly-Race Lemma (§3.3) cannot be discharged by surrender — only by the victim's elimination.*

3.3 The Deadly-Race Lemma

Lemma 1 (Deadly-Race Lemma)**.** *Let a custody scheme admit a* gray *coercion outcome (Definition 3): on release, $\mathcal{A}$ holds a feasible path to K ($p_A > 0$) while $\mathcal{V}$ retains one too ($p_A < 1$). Then the priced adversary* $\mathrm{WDY}[W, c]$ *(Definition 4) has a **strictly positive incentive to eliminate** $\mathcal{V}$ (and any co-racing heirs) whenever $(1 - p_A)\,W > c$.*

Proof. With $\mathcal{V}$ alive and free, $\mathcal{A}$'s expected take from the race is $p_A\,W$. By NNPP (Assumption 4), $\mathcal{A}$ *cannot* instead obtain or demand a verified assurance that $\mathcal{V}$ will not race — no such proof exists — so the *only* instrument available to $\mathcal{A}$ for pushing the win probability toward 1 is to remove the competing racer. Eliminating $\mathcal{V}$ drives $\mathcal{A}$'s win probability to ≈ 1 and the expected take to $\approx W$. The marginal gain is $W - p_A W = (1 - p_A)\,W$, and a rational $\mathcal{A}$ acts on it exactly when it exceeds the cost c. In any gray outcome $p_A < 1$, so $(1 - p_A)\,W > 0$; for assets large relative to c the inequality holds. □

Three remarks fix the result's meaning and scope:

Remark 6 (It is decision-theoretic, not cryptographic)*.* Lemma 1 says nothing about breaking a cipher; it holds *whatever* the underlying cryptography, because it lives in the incentive layer threat models omit. This is why "the crypto is strong" is no defense — the danger is created by the *shape* of the recovery path, not its hardness.

Remark 7 (Release is the trigger)*.* The incentive is a property of the released gray state: it is $\mathcal{V}$-as-free-racer whom $\mathcal{A}$ is paid to remove. A scheme that simply hands $\mathcal{A}$ the funds outright kills no one; the gray fallback is *worse than* that clean loss. And by Corollary 1 no amount of surrender removes the incentive — $\mathcal{A}$ can never confirm $\mathcal{V}$ kept no resumable copy — so only the victim's elimination closes it.

Remark 8 (The danger peaks where the marketing promises safety)*.* A "graceful" fallback — a slow-but-feasible recovery left on a seized device, a time-locked rescue, a delayed vault — holds p_A well below 1 while leaving W fully at stake, i.e. it *maximizes* $(1 - p_A)\,W$. Designs that degrade gracefully thus tend to sit squarely in the region of **largest** assassination incentive. Graceful degradation, a virtue everywhere else, is here a liability — hence the title.

3.4 The No-Gray-Area Criterion

Criterion 1 (No-Gray-Area Criterion)**.** *Under a fixed threat model, a coercion-*safe *scheme must admit **no gray outcome**: every encounter must resolve to a* clear *endpoint — $p_A = 0$ (the attack plainly* fails*: Σ feasibly reaches nothing and $\mathcal{V}$ holds nothing transmissible left to surrender) or*

$p_A = 1$ *(it plainly* succeeds*: $\mathcal{A}$ obtains K within the encounter, so no post-release race exists). The forbidden interval $0 < p_A < 1$ is exactly where Lemma 1's incentive lives.*

The Criterion is a **bar, not a construction**: it fixes what a design must achieve (collapse every outcome out of the gray interval), not how. The two clear endpoints are not equally desirable to $\mathcal{V}$ — $p_A = 1$ costs $\mathcal{V}$ the funds — but both are *survivable*: neither leaves a released racer whom $\mathcal{A}$ profits by killing. Coercion resistance, properly posed, is therefore a bar on **the holder's safety** first, and only then on the funds.

The Criterion as a theorem

The Criterion need not stay a stipulation. Cast the encounter as a one-move game and "coercion-safe" as a predicate on it, and no-gray becomes exactly the safety condition.

Definition 5 (Coercion game and coercion-safety)**.** Fix a scheme Σ and a priced adversary WDY$[W, c]$ (Definition 4). The game $\textbf{Coerce}_\Sigma$ is: (1) $\mathcal{A}$ mounts an encounter — device seizure (D3–D7) and the bounded coercion oracle (D3) — reaching a post-encounter state with a set R of *racers*: the parties that can still reach K within feasibility and act on it (spend, or rescue/cancel to defeat $\mathcal{A}$'s spend) — the released $\mathcal{V}$, and any remote delegate. Write $p_A(S)$ for $\mathcal{A}$'s probability of prevailing against racer set S (monotone: fewer racers, weakly higher p_A); the scalar p_A of Definition 2 is $p_A(R)$. (2) $\mathcal{A}$ picks $a \in \{\mathsf{spare}, \mathsf{kill}\}$, where kill removes $\mathcal{V}$ from R — by reachability $\mathcal{A}$ can eliminate only racers it can *reach*, never a remote delegate. (3) the race resolves, with payoff

$$U_{\mathcal{A}}(a) \;=\; p_A(S_a) \cdot W \;-\; c \cdot \mathbf{1}[a = \mathsf{kill}], \qquad S_{\mathsf{spare}} = R, \quad S_{\mathsf{kill}} = R \setminus \{\mathcal{V}\}.$$

Σ is **coercion-safe** if no WDY$[W, c]$, at any reachable state, strictly prefers kill.

Theorem 1 (No-Gray-Area, as a theorem)**.** *Write $\pi := p_A(R \setminus \{\mathcal{V}\}) - p_A(R) \geq 0$ for $\mathcal{V}$'s* pivotality *— the win-probability $\mathcal{A}$ gains by removing $\mathcal{V}$ as a racer. Σ is coercion-safe (Definition 5) against every* WDY$[W, c]$ *if and only if $\pi = 0$ at every reachable state: $\mathcal{V}$ is never pivotal.*

Proof. The kill increment is $\Delta U := U_{\mathcal{A}}(\mathsf{kill}) - U_{\mathcal{A}}(\mathsf{spare}) = \pi W - c$. ($\Leftarrow$) If $\pi = 0$ at every reachable state then $\Delta U = -c < 0$: no WDY$[W, c]$ ever prefers kill, so Σ is safe. ($\Rightarrow$, contrapositive) If $\pi > 0$ at some reachable state then $\Delta U = \pi W - c > 0$ for any (W, c) with $W > c/\pi$; that priced adversary exists and strictly prefers kill, so Σ is unsafe. □

A game-based (experiment-and-advantage) restatement of this equivalence — with the adversary's clauses D0–D8 as oracles and *reaching a gray state* as the winning event — is given in Appendix A; there coercion-safety is exactly $\Pr[\text{win}] = 0$.
Pivotality $\pi = 0$ has exactly the two clearance routes of Corollary 2. **R1 (individual custody, $R = \{\mathcal{V}\}$):** removing $\mathcal{V}$ leaves no racer, so $p_A(\emptyset) = \mathbf{1}[p_A > 0]$ and $\pi = \mathbf{1}[p_A > 0] - p_A$, which is 0 *iff* $p_A \in \{0, 1\}$ — the "no gray outcome" of Definition 3, recovered as the special case. **R2 (shared custody, a remote delegate $D \in R$):** killing $\mathcal{V}$ leaves D, so $\pi = 0$ *iff D dominates $\mathcal{V}$ as a racer* — D prevails against $\mathcal{A}$ whenever $\mathcal{V}$ would, so $\mathcal{V}$ is never the unique blocker. Here the state may be *gray* ($0 < p_A(R) < 1$) yet safe, because the pivotal racer is unreachable; this is what it means for R2 to clear the Criterion without removing the race. It is contingent: if D is co-located (hence reachable — $\mathcal{A}$ removes it too) or merely *slower* than $\mathcal{V}$ (so $\mathcal{V}$ is sometimes the unique blocker), then $\pi > 0$ and the assassination incentive returns — exactly the "escape closes" failure and the BitVault case (§5), where the "delegate" is $\mathcal{V}$'s own alert wallet and $\mathcal{V}$ is maximally pivotal.

Criterion 1 is thus a *theorem*, and pivotality unifies it with the reachability corollary: coercion-safety is *$\mathcal{V}$-never-pivotal*, of which "no gray outcome" is the individual-custody reading and "gray but the pivotal racer is out of reach" is the shared-custody one. What the single-shot model leaves open is the *utility* behind ΔU (repeated encounters, partial extraction, heir cascades, $\mathcal{A}$'s risk-aversion); sharpening that (§3.5, "still to formalize") is the remaining work, not the equivalence.

Grading protocols: naive, weak, strong

With coercion-safety in hand we can *grade* a custody protocol by how much of the adversary it withstands — from mere obscurity, through secrecy-without-safety, to the full bar. The three levels below are the vocabulary §5 classifies existing designs with.

Definition 6 (Naively coercion-resistant custody protocol)**.** A protocol is **naively coercion-resistant** if it *may* resist a *sub-Kerckhoffs* attacker — one with the physical capabilities **D1–D7** but lacking Kerckhoffs knowledge (**D0**, Assumption 2) — because denial, decoy, or obscurity *may* work when the attacker does not know $\mathcal{V}$'s setup. Both hedges are load-bearing: the guarantee is contingent on the attacker's *ignorance*, not on any structural property, so it offers *no* guarantee against the full WDY adversary. Examples: plausible-denial and decoy wallets, or time-gated transactions cancelable through an *obscure* key — each collapses the moment D0 is granted.

Definition 7 (Weakly coercion-resistant / weakly WDY$[W, c, b, t]$-resistant)**.** A protocol is **weakly coercion-resistant** (equivalently, **weakly** WDY$[W, c, b, t]$**-resistant**) if a single encounter with the full WDY adversary (**D0–D8**) does not, *by itself alone*, yield the master secret K, but *may* leave a *deadly-race* state: a reachable state of positive pivotality $\pi > 0$ (Theorem 1) in which $\mathcal{A}$ holds a feasible-but-slower path and $\mathcal{V}$ remains a reachable racer. The secret is withheld *during coercion*; the holder is not safe: *after coercion finishes* $\mathcal{A}$ may enact deadly race and ultimately seize K nevertheless.

Definition 8 (Strongly coercion-resistant / strongly WDY$[W, c, b, t]$-resistant)**.** A protocol is **strongly coercion-resistant** (equivalently, **strongly** WDY$[W, c, b, t]$**-resistant**) if a single encounter with the full WDY adversary (**D0–D8**) yields *neither* K *nor* a deadly-race state — every reachable state has $\pi = 0$. This is exactly coercion-safety (Definition 5, Theorem 1): the bar Criterion 1 sets, and the target the construction of §4 meets.

Definition 9 (Time-gating custody protocol)**.** A custody protocol is **time-gating** if reaching its master secret K requires completing a *time-gate* — a computation or delay of wall-clock cost G (a memory-hard derivation, or a time-lock) — so that K is unavailable before G elapses; G is the design parameter, set against the adversary's coercion-time budget t (D3). Time-gating is a *mechanism*, not a strength: a given time-gated protocol instantiates one of the three levels above according to the outcome it forces — **naive** if it relies on an obscure abort or cancel key (safe only against a sub-Kerckhoffs attacker); **weak** if the gate leaves a feasible-but-slower path, keeping $\pi > 0$; **strong** if every reachable state has $\pi = 0$ (Theorem 1), no matter the mechanism. It is the running example of the taxonomy.

3.5 Reachability corollary — the two clearance routes

Corollary 2 (Reachability)**.** *Lemma 1's incentive is realizable only against a racer $\mathcal{A}$ can* reach*. A scheme can therefore satisfy Criterion 1 in exactly two ways:*

- **(R1)** remove the race — *arrange that no feasible seizure path exists, forcing $p_A = 0$ (custody can stay* individual*); or*
- **(R2)** remove $\mathcal{A}$'s reach over the winner — *make the party who* will *win the race a* remote delegate *beyond $\mathcal{A}$'s coercive reach (custody becomes* shared*, at the cost of a trusted party).*

R1 and R2 are the classification axis of §5 — *leave no race to run* vs. *put the racer out of reach.* TGPO (§4) takes R1; time-lock-rescue vaults take R2.

Remark 9 (The honest cost of the remote-delegate escape). It is tempting to say NNPP (Assumption 4) also weakens R2, on the grounds that $\mathcal{V}$ cannot prove that no local, coercible copy of the delegate's rescue capability exists. It does not. By Theorem 1 the holder-safety of R2 is exactly $\pi = 0$, which holds *iff* D access-dominates $\mathcal{V}$ (Definition 10); a hidden backup changes nothing, because **assassination removes a racer but never extracts a secret**. Were such a copy extractable, $\mathcal{A}$ would take it *during* the encounter — a clean success ($p_A = 1$, no post-release race), the survivable endpoint; were it not, D still dominates and killing $\mathcal{V}$ is pointless. Neither branch is a deadly race. NNPP bites only in the *original* race (Lemma 1), where $\mathcal{A}$ *holds* a feasible-but-slower path from the seized device and cannot be reassured that $\mathcal{V}$ will not race; it does not reach the delegated case.

One honest residual difference survives, and it is about *funds*, not the holder. The delegate's capability is *transmissible*, so R2 keeps a clean-loss exposure that R1 lacks: if $\mathcal{V}$ *did* retain a coercible copy, coercion extracts it and the funds are lost cleanly ($p_A = 1$). R1's secret is *non-transmissible* (D6–D7, Assumption 3), so full coercion extracts nothing.

The decisive objection, though, is an elephant in the room: an access-dominating delegate *is not self-custody.* Its entire clearance rests on assuming D will **refuse a legitimate owner pleading for their life** (D8) — the very refusal that defeats $\mathcal{A}$'s coerced plea also defeats $\mathcal{V}$'s genuine one. Whether D's state is *sufficient* for K or merely *necessary*, shared custody reinstates exactly the trusted counterparty the design set out to remove. *Not your keys, not your coins*: that, more than any impossibility, is the standing case for the individual-custody route (R1).

Remark 10 (Still to formalize). The framework-side elevation is done: Theorem 1 makes Criterion 1 a theorem (coercion-safe iff the victim is never pivotal, of which no-gray is the individual-custody case) in the single-shot model, and NNPP is assumed and justified in §3.2. **(i)** What remains there is a *richer utility model* behind the kill increment — repeated encounters, partial extraction, heir cascades, $\mathcal{A}$'s risk-aversion (and, optionally, a fully asymptotic PPT-with-oracle restatement) — which would sharpen Lemma 1 and Theorem 1 beyond the single shot.

On the construction side, one scoping note. Theorem 3 discharges $p_A = 0$ *for every seizable state except the residual window* by case analysis under the feasibility line (S1/S2, Stipulation 1) — a standard *key-size* stipulation, not a reduction to a named hard problem, and none is needed for a key-sizing argument (§8). The residual window is *not* $p_A = 0$: it is a bounded $p_A > 0$ exception argued *incentive*-safe against WDY$[W, c]$ (§4.7), a "more likely than not" statement rather than a theorem. **(ii)** Turning that residual argument into a *quantitative bound* is the remaining construction-side item; the softest premise of all — that a transmissible-resistant perceptual oracle exists at all — is empirical, and flagged in §8.

3.6 Geographic, delegated, or tacit — what forces TKBA

Take a *time-gating custody protocol* (Definition 9) and ask *how* it can attain the $\pi = 0$ of Theorem 1 against a single WDY encounter (one *site*, D4). There turn out to be exactly three mechanisms, and each spends a different one of Great Wall's *Four Properties* [9]:

(1) knowledge-based authentication — the secret in memory alone, i.e. *non-geographic*;
(2) individual custody;
(3) non-obscurity — which *is* the WDY Kerckhoffs premise (D0);
(4) coercion resistance.

Definition 10 (Access-dominating delegate)**.** For a party P let $\ell_P(\tau)$ be its latency to reach and act on K (spend, or rescue/cancel) from the state at time τ. A delegate D is **access-dominating** if $\ell_D(\tau) \le \ell_V(\tau)$ at every τ — K is, at all times, at least as available to D as to the owner. Then D is never slower than $\mathcal{V}$, so removing $\mathcal{V}$ leaves the race unchanged, $p_A(R \setminus \{\mathcal{V}\}) = p_A(R)$, giving $\pi = 0$ (route R2). Operationally the owner *periodically syncs* D's state, holding $\ell_D \le \ell_V$.

Remark 11 (Dominance can be made checkable). The sync is best run during *peacetime* (the quiescent interval between encounters), and it can be made *verifiable*: two time-separated digests of D's state reveal not only that D leads but its *rate* v_D. Knowing its own rate v_A and the two remaining distances to K — Δ_{S_A} for $\mathcal{A}$, Δ_{S_D} for D — $\mathcal{A}$ can settle the race in advance: it can win only if $\Delta_{S_A}/v_A < \Delta_{S_D}/v_D$. A delegate running near the frequency ceiling (D2) against a slower $\mathcal{A}$ makes racing plainly futile — turning D's dominance from $\mathcal{A}$'s *credence* into something $\mathcal{A}$ can *check*. Whether D's held state is *sufficient* for K or merely *necessary* (so that defeating $\mathcal{A}$ needs only that D can *block*, not spend) does not change the argument.

Definition 11 (Geographic protocol)**.** A protocol is **geographic** if K's gating is partitioned across ≥ 1 *sites*, so that reaching K requires the co-located seizure and coercion of co-located principals (D3–D7) to be deployed at *every* such site. A single WDY encounter is confined to one site (D4); the moment K's gating requires a site the encounter does not occupy, it has no feasible path, $p_A(\varnothing) = 0$, so $\pi = 0$ (route R1). Custody may stay individual (the other sites are the owner's own devices), but K now *requires geographic locations* — which is exactly the negation of Property 1 (knowledge-based ≡ non-geographic).

Remark 12 (The geographic route is legitimate but inherently physical)*.* Definition 11 clears the bar honestly; but it does so by moving the burden onto *physical* distance and guarding, and an honest account should name the avenues it leaves $\mathcal{A}$ rather than gloss over them. Against a protocol whose K needs ≥ 1 site of custody, $\mathcal{A}$ may — singly or in combination:

1. **catch $\mathcal{V}$ *at* a site**: then the encounter's single site (D4) *is* a gating site, and only the *other* sites stay out of reach — which is why the definition needs a site the encounter does not occupy, and why a lone site is the weakest case;
2. **burgle the sites without $\mathcal{V}$**, taking the tokens directly — possibly after enacting a deadly race, since removing $\mathcal{V}$ (Lemma 1) buys uncontested time to travel and break in;
3. **abduct $\mathcal{V}$ and move them from site to site**, seizing each in turn — a chain of single-site encounters, bounded only by travel time and the coercion-time budget t (D3);
4. **coerce $\mathcal{V}$ into having allies retrieve and transmit** the remotely-kept tokens — excluded here by D8 (the remote party does not comply), but named for completeness.

None of these is defeated by *cryptography*: each is bounded, if at all, only by *physical* facts — how far apart the sites sit, how well they are guarded, how much travel the window t forbids. A geographic protocol is thus a *legitimate but inherently physical* clearance, and one that *scales in cost*: more sites, farther apart and better defended, cost more to hold and to guard. Its safety is bought in physical, not cryptographic, coin — the reduction the next two remarks trace, and that the tacit route (R1) exists to avoid.

Remark 13 (Literal Race)*.* A geographic protocol (Definition 11) can turn the deadly race into a *literal* one: once the wrench attack ceases, whichever party — $\mathcal{V}$ or $\mathcal{A}$ — first roams to and reaps enough of the remote custody sites takes the prize W. The race is now run over physical ground rather than computation. This exposes a second, purely *physical* lever: depending on how well those remote sites are defended, $\mathcal{A}$ may assess its probability of successfully seizing the remote secrets — hence its pivotality π — as so low that $\pi W \leq c$, i.e. $W \leq c/\pi$; by Theorem 1 the (literal) deadly race is then *not worth enacting.*

Remark 14 (The geographic escape reduces to physical security)*.* But note what that concedes: security now rests on the question *"how effective are the physical defenses against the assumed physical attack capability?"* — Bitcoin reduced to a *glorified gold* guarded in vaults (cf. the Justification document [5]). The tacit route (R1, Definition 12) is precisely what escapes this regress: it reduces the physical-security problem to an *actual cryptographic* one, making banking-grade physically-secure self-custody available at the cost of running a piece of free open source software.

Remark 15 (Cyber-physical race)*.* A protocol that admits *two or more* feasible paths to K — one running through the collection of physically distributed tokens and another that does not (say an undisclosed, locally hidden shortcut, or a state seized *during* the wrench attack) — yields a *cyber-physical* race: one racer works to complete a KDF (or time-lock) computation while the other roams the custody sites. The temporal and spatial clearance routes are then run in parallel by different parties, and p_A is fixed by whichever path is faster — a mixed regime that the pure-temporal (tacit) and pure-spatial (geographic) analyses each capture only half of.

Definition 12 (Tacit gating (TKBA))**.** A protocol is **tacitly gated** if it attains coercion resistance by gating K behind a *non-transmissible* in-head input (Assumption 3): the deep perceptual stages are recognizable but not dictatable. A single encounter can then neither extract the input (non-transmissible, and the coercion-time budget t of D3 is too short to walk the derivation) nor reconstruct it offline (the seized state meets the floor), so $p_A(\varnothing) = 0$ and $\pi = 0$ (route R1) — now *at a single site, with the secret in memory only.* This keeps Properties 1 *and* 2. In TGPO the gate is *temporal* — a memory-hard chain with $G > t$ (§4).

Remark 16 (Geography-gated perceptual oracle). The perceptual gate need not be *temporal.* One could equally conceive a *geography-gated* perceptual oracle: instead of the deep stages being gated by a long hashing chain or a TLP, they sit at a physically protected custody site. A wrench attack then succeeds only if it is mounted *at that protected site, while* $\mathcal{V}$ *is present.* This is not what TKBA was conceived for — it trades Property 1 (non-geography) for a spatial gate — but it is a genuinely valid alternative that some users may prefer, and the trichotomy (Theorem 2) already covers it as the *geographic* leg carrying a perceptual oracle.

Remark 17 (Remote-delegate-gated perceptual oracle). Analogously, the perceptual gate can be held by a *remote delegate*: the deep stages complete only with a party beyond $\mathcal{A}$'s reach, sacrificing *individual custody* (Property 2) rather than non-geography (Property 1). As with the geographic variant, this is not what TKBA was conceived for, but it may still be valid. One qualification is essential: its efficacy is contingent on clause D8 — $\mathcal{A}$ cannot use $\mathcal{V}$ to remotely coerce the delegate into cooperating. That requires either trusting a remote agent, or keeping physical distance from an agent one already trusts, lest $\mathcal{A}$ reach both at once (see Remark 28).

Remark 18 (Obscure perceptual oracle — the invalid alternative). The same move fails for *obscurity.* Gating the perceptual oracle on knowledge of the protocol alone — merely hoping $\mathcal{A}$ lacks it — makes no sense, by a dilemma. If $\mathcal{A}$'s ignorance could be taken for granted, then simpler obscure schemes (denial, decoy, or an obscure abort) would already suffice, and the perceptual oracle buys nothing. If it cannot — i.e. under Kerckhoffs (D0) — and the only thing gating the perceptual parameters is knowledge of their existence, then by definition they are *available* at the attack state; hence by D6 and D7 they, and their respective points, are seized — utterly ineffective. Unlike the geographic and remote-delegate variants, there is no regime in which this one clears the bar.

Bottom line. The whole premise of TKBA is to trade *availability* for coercion resistance: it lowers the likelihood that K is seizable at any given moment, at the cost of making K less available to the legitimate user — while still not yielding a deadly race. Kerckhoffs throws obscurity out of the window, so the only ways to restrict that availability are *geographically*, through *remote delegates*, or through *cryptographic time-gating* (TGPO). These are exactly the three DRL-safe routes of Theorem 2; the perceptual oracle can ride either of the first two but is native to the third.

Theorem 2 (DRL-safety trichotomy)**.** *A time-gating custody protocol (Definition 9) is DRL-safe (Theorem 1) against a WDY adversary only if, at every reachable state, it is* ***access-dominating-delegated*** *(R2, Definition 10),* ***geographic*** *(R1, Definition 11), or* ***tacitly gated*** *(R1, Definition 12).*

Proof sketch. By Theorem 1 DRL-safety is $\pi = 0$, i.e. $p_A(R \setminus \{\mathcal{V}\}) = p_A(R)$. Either $R \setminus \{\mathcal{V}\}$ retains a racer at least as fast as $\mathcal{V}$ — an access-dominating delegate (Definition 10) — or it does not, whence (dropping the degenerate clear-success $p_A(R) = 1$) $p_A(\varnothing) = 0$: a single encounter has *no* feasible path, so a piece of K's gating is withheld from an adversary that seizes every co-located device and coerces the present owner. A co-located WDY has only two blind spots — *elsewhere* (another site: geographic) and *the unextractable mind* (non-transmissible, behind a time-gate exceeding the window: tacit). No third withholding exists. □

Corollary 3 (TKBA characterizes the Four Properties)**.** ***Necessity.*** *Property 1 (knowledge-based, non-geographic) demands a store of* maximal availability *— one that travels with the owner, not a fixed location [9]. Several qualify: the brain, but equally a body-implanted chip, a biometric token, or a USB stick carried everywhere. Property 4 (coercion resistance against a*

WDY) prunes them: every carried device *is co-located at the attack site, hence seized (D4), and a biometric is coerced — only the* brain*, the one store that is not a device, survives. The brain still holds both coercible (explicit, memorizable) and non-coercible content, so the same Property 4, now against the coercion oracle (D5), forces the secret to be the latter:* non-transmissible*, i.e. tacit. Non-geography narrows the store to portable ones;* non-seizability *(Property 4) picks the brain among them and makes its content tacit. Thus* (1) ∧ (3) ∧ (4) *yield a tacit brain secret, and* (2) *that it is the owner's alone — TKBA.* ***Sufficiency*** *is the reverse check — TKBA's input is in memory (1), needs no second party (2), resists a protocol-aware WDY by non-transmissibility rather than a hidden trick (3), and forces $p_A(\varnothing) = 0$ ($\pi = 0$, up to the tolerated residual) (4). Hence*

$$\textit{TKBA} \iff (1) \wedge (2) \wedge (3) \wedge (4).$$

TGPO (§4) is that construction — neither geographic nor delegated, yet DRL-safe. Of the three DRL-safe routes (Theorem 2), the delegate forfeits (2) and the geographic route forfeits (1); only the tacit leg keeps all four.

This characterization needs *no* exhaustiveness argument: Property 1 narrows the store to portable ones, and Property 4 pins it to the brain (the sole non-seizable one) and its content to non-transmissible. The *general* trichotomy (Theorem 2), which classifies *every* DRL-safe time-gating protocol, does rest on a two-blind-spots enumeration (a piece denied a co-located WDY is either *absent* — another site, geographic — or *present but unextractable* — in-head and non-transmissible, tacit); the Four-Properties equivalence stands independently of that step.

3.7 A fifth property — no hostage token

Devicelessness (Property 1) yields a fifth security property the Four have so far left implicit: immunity to *indirect* coercion. Wherever a *necessary* token is stored outside the holder's head — on a carried device or at a site of custody — that token is a *hostage*. Even where seizing K itself is judged infeasible (or not worth the risk), $\mathcal{A}$ can seize the token, or threaten its destruction, and trade it for something else entirely: the *formal, irrevocable* transfer of another good or service. The property is orthogonal to the remote-delegate route (R2) but, by construction, *denied* to any geographic protocol (Definition 11), which needs precisely such external tokens; only the deviceless, in-head secret has nothing to take hostage.

Remark 19 (Indirect wrench via a hostage token). Even a device-dependent setup whose K is deemed unwrenchable can present a surface for an *indirect* wrench attack through its most wrench-vulnerable device-stored token. $\mathcal{A}$ seizes that token and holds it hostage, to be traded for the formal transfer of another of $\mathcal{V}$'s assets — possibly one *less* liquid and *more* traceable, and so otherwise easier to rescue: a car, jewellery, real estate. For instance, take a setup in which $\mathcal{V}$ carries one necessary token at all times and keeps the other at a site beyond $\mathcal{A}$'s wrench reach. Caught *away* from that site, $\mathcal{V}$ can be relieved of the carried token and threatened with its destruction unless they *legally* — hence irrevocably — transfer some otherwise-rescuable asset. And a site of custody, not personally guarded by its owner, is itself a stationary (and possibly unguarded) surface for token seizure.

Remark 20 (Weaponizing the hostage to probe for backups). The same lever can be turned to *detect* what NNPP (Assumption 4) holds $\mathcal{V}$ cannot disprove. By how dearly $\mathcal{V}$ ransoms a hostage token, $\mathcal{A}$ reads off information about backups: a high rescue price paid suggests no backup to that token exists. This trades NNPP's clean simplification for a game-theoretic can of worms — either party may bluff — and lies beyond the scope of this work.

Remark 21 (Deviceless TKBA avoids the can of worms). Deviceless, classic TKBA has no external token to take hostage, and so escapes the whole tangle: with the secret in the head alone, there is nothing to seize, ransom, or read a backup off.

Remark 22 (Denial is inherently plausible). NNPP (Assumption 4) leaves $\mathcal{A}$ unable to disprove an undisclosed shortcut backup — but for the deviceless holder that cuts the other way. Under

Kerckhoffs $\mathcal{A}$ knows the protocol, and so knows a TGPO setup reaches K from the head alone: $\mathcal{V}$ has no *need* of a shortcut device. A denial — "there is no backup" — is therefore *consistent with a perfectly ordinary setup*, not a suspicious claim to be tortured out of. Where a device-dependent holder might plausibly keep one, the deviceless holder plausibly keeps none: the very absence of need is what makes the denial more credible, and consequently, $\mathcal{A}$ more inclined to desist.

4 Construction — the Time-Gated Perceptual Oracle (clears by removing the race)

We now exhibit a construction that clears Criterion 1 by the first route of Corollary 2 — the *tacit* route that Corollary 3 shows is forced by holding all Four Properties at once: it admits *no feasible seizure path*, so custody remains individual. We call it the **Time-Gated Perceptual Oracle** (TGPO) — *time-gated* for the memory-hard orbit and time-lock puzzle, *perceptual* for the tacit fractal recognition, *oracle* for its evaluate-only-by-the-holder semantics; it is the coercion-resistance protocol at the core of the Great Wall / Great Wallet self-custody application. We give here the parts the theorem needs; the concrete instantiation parameters — the memory-hard cost tiers, iteration factoring, and the timechain salt — are specified in the Great Wall Core design document [10].

4.1 The orbit

Let H be a cheap, entropy-preserving one-way hash and H^* a memory-hard (Argon2-class) hash of tunable cost. Let σ be a public, precomputation-resistant *timechain salt* whose cross-user amortization is negligible (the companion Namtso primitive [11]). Define an **orbit** of N deep stages (Figure 2):

$$o_0 = \mathsf{H}(\sigma), \tag{1}$$

$$u_i = \mathsf{H}(o_i \parallel Sh_i), \qquad 0 \le i \le N-1, \tag{2}$$

$$o_i = \mathsf{H}^*(\, u_{i-1}\,), \qquad 1 \le i \le N, \tag{3}$$

$$K_i = \mathsf{H}(o_i \parallel Sh_i), \qquad 0 \le i \le N, \tag{4}$$

where the raw inputs $\{o_{i-1}, Sh_{i-1}\}$ are **wiped** in the instant between the cheap inner hash and the long memory-hard step. Computing $u_{i-1} := \mathsf{H}(o_{i-1} \| Sh_{i-1})$ first (instant), wiping, then running the long $o_i := \mathsf{H}^*(u_{i-1})$, means the raw prior orbit point and Shamir secret exist only momentarily — not across the whole memory-hard window. This directly shrinks the window in which $\{o_{i-1}, Sh_{i-1}\}$ is seizable (D4), and is the reason not to use $\mathsf{H}^*(o_{i-1} \| Sh_{i-1})$ directly.

On notation. H and H^* are *generic* designations — any suitable cheap, and memory- and depth-hard (Argon2-class), instantiation serves. Note that u_i and K_i share the formula $\mathsf{H}(o_i \| Sh_i)$; they are nonetheless *distinct* objects, separated by *role* — u_i is consumed by the next memory-hard step, whereas K_i is exposed as a usable master secret. A deployment keeps the two apart by domain separation, a specification detail beyond our scope.

4.2 No single-point oracle

Each stage i carries not an individual point but a *whole Shamir polynomial* [12]. Stage i has s_i points $p_{i,j}$, each a 32-bit leaf-area of a fractal $\theta_{i,j} = \mathsf{H}(o_i \| j)$; any r_i of them reconstruct the degree-$(r_i{-}1)$ polynomial Sh_i of $r_i \cdot 32 \ge 64$ bits ($r_i \ge 2$). The orbit absorbs the *full* polynomial Sh_i, not its constant term $f(0)$.

Remark 23 (Block-orbit $r \ge 2$ condition). Because the smallest object any orbit link commits is a full stage (≥ 64 bits), no individual point $p_{i,j}$ is ever checkable in isolation — there is no sub-threshold oracle against which a single guessed point could be verified. This is the block-orbit $r \ge 2$ condition, realized as Shamir $r_i \ge 2$ per stage.

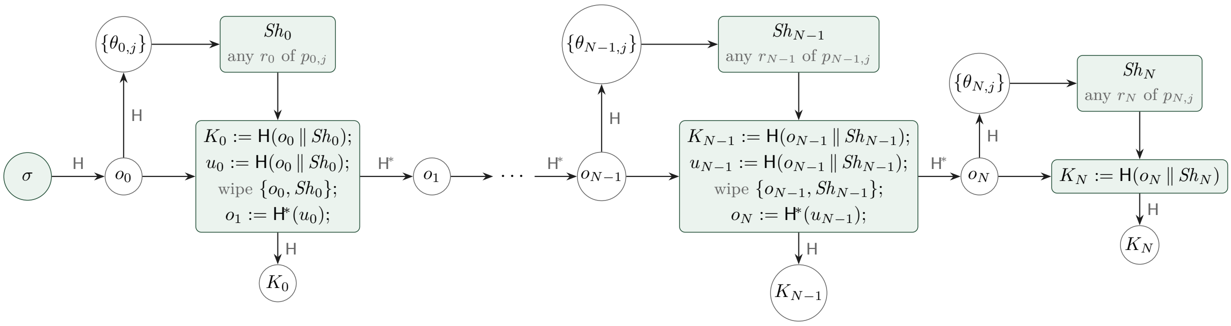

Figure 2: **The TGPO orbit** (§4). A public, precomputation-resistant timechain salt σ (the Namtso primitive) seeds a chain of N memory-hard stages. At each stage the raw inputs $o_{i-1} \,\|\, Sh_{i-1}$ are folded by the *cheap* inner hash H and *wiped* before the long memory-hard step H^*, so seizable material exists only for an instant — shrinking the window in which o_{i-1}, Sh_{i-1} are reachable. Each board o_i derives the fractal parameters $\theta_{i,j} = \mathsf{H}(o_i \| j)$ on which its points $p_{i,j}$ are recognized; any r_i of those points reconstruct the whole Shamir polynomial Sh_i, absorbed into the *next* orbit link, so no single point is checkable in isolation. The terminal secrets $K_i = \mathsf{H}(o_i \,\|\, Sh_i)$ deliberately use the cheap hash, keeping the final window instant. By construction, coercion resistance happens for $i \geq 1$.

4.3 Tacit multi-fractal stages

The points $p_{i,j}$ instantiate Assumption 3: they are recognizable on their fractal boards but not transmissible without $\theta_{i,j}$ (TKBA_1), and the deep $\theta_{i,j}$ are regenerated from o_i — which itself requires walking the memory-hard chain (TKBA_2). Using several shallow-zoom fractals per stage lets a stage carry $r_i \cdot 32$ bits at low zoom depth per board, keeping each point recognizable while tuning stage entropy in 32-bit steps.

4.4 Cheap terminal secret

The coercion-resistant master secrets are all

$$K_i, \qquad 1 \leq i, \tag{5}$$

deliberately using the *cheap* $K_i = \mathsf{H}(o_i \,\|\, Sh_i)$.

Each K_i ($i \geq 1$) is a full instantiation of the abstract master secret K that §3 reasons about: the orbit yields not one secret but a nested *family* of master secrets, and every K-result — the floor (Theorem 3), the deadly-race analysis — holds of each. K_0 is set aside: stage 0's board o_0 derives from the public salt alone ((4)), so it sits below the memory-hard chain that gives the deeper stages their floor.

A memory-hard last step would only *prolong* the window (§4.6) in which o_N and the final points are live (paying an H^* *after* the last point is recognized), worsening the very window we shrink in (4). The resistance therefore lives in *entropy*: standard deep stages carry 96 bits ($r_i = 3$), so a cheap ledger brute of Sh_N from a bare o_N costs 2^{96} — infeasible by S3 — while the terminal window stays instant.

4.5 The floor

Theorem 3 (The floor on the master secret)**.** *Under Assumptions 2–3 and the standard build of* (4)–(5)*, from every seizable state* save a single residual window *(§4.6) a target master secret K_N is* prohibitive *to reach — its cost meets the floor (Stipulation 1). Consequently a device seizure outside that window yields nothing feasible — a* clear attacker failure *in the sense of Criterion 1, with no racer for $\mathcal{A}$ to reach.*

Proof by case analysis over seizable states. From a seized o_i $(i < N)$, reaching K_N requires guessing $Sh_i, \ldots, Sh_N$; each Sh_l guess is $2^{r_l \cdot 32}$ and is H^*-gated, since it is checked only by recomputing the memory-hard link (4). Preimage resistance of H, H^* makes the orbit forward-only: a seized o_i cannot yield o_{i-1}. We enumerate:

- *Pure device seizure* (no coercion; only time-lock-gated [13] blobs at rest). All tacit points are required; K is unreachable.
- *Wrench of stage 0* ($p_{0,\cdot}$, always present since σ is public). $\mathcal{A}$ computes o_1 with one H^*; from o_1, K_N costs $2^{\sum_{j \geq 1} r_j \cdot 32}$, which is 2^{96} or more in the recommended standard builds [10], each guess H^*-gated — prohibitive.
- *Mid-orbit seizure* at o_i $(0 < i < N)$. K_N costs $\geq 2^{96}$ cheap and is H^*-gated per guess — prohibitive.
- *Residual* o_N present. Treated separately in §4.6.

Every case except the residual leaves K_N at the floor (Stipulation 1). □

4.6 The tolerated residual seizure window

One sliver escapes the floor, and it is first a *seizability* gap — not yet a deadly race. It is the lone exception of Theorem 3: a wrench that lands with the final board o_N imminently or currently live, and subdues $\mathcal{V}$ before the wipe ((4)), catches the derivation essentially complete and $\mathcal{V}$ under control. In that coincidence $\mathcal{A}$ *does* reach K and the stash is lost — we concede it plainly. The floor is a claim about *seizability*, and precisely here it fails; what keeps the failure a mere *residual* is how narrow the coincidence is — a precise landing on the brief live-o_N interval, and a subdual before any wipe or reset.

4.7 The tolerated residual deadly-race window

Funds-loss, though, is a *separate* axis from the *deadly race*, and it is the latter this section turns on: does the same window also manufacture an incentive to *kill* $\mathcal{V}$? That does *not* follow from the seizure. Killing buys $\mathcal{A}$ only a *shaving* of the final step — after the *second-to-last* point of stage N is surrendered, a cheap $2^{32} \cdot \mathsf{H}$ brute ((5) is cheap) reaches K in $\approx$1 minute, versus waiting for a cooperating coerced $\mathcal{V}$ to recollect it. Whether that shaved minute pays an assassination takes *further* conditions, which we test against the priced adversary WDY$[W, c]$ (Definition 4), the model appropriate to a quantitative window — against the only thing that matters: the kill condition of Lemma 1, $(1 - p_A)\,W > c$. We argue it is, more likely than not, unmet.

Two events govern the marginal value of killing $\mathcal{V}$ in this window:

- Corr — *the points* $\mathcal{V}$ *has surrendered so far are correct*; and
- Esc — *the* $\approx$1–2 *minutes shaved by brute-forcing the last one or two points, rather than waiting for a cooperating coerced* $\mathcal{V}$ *to recollect them, make the difference between a successful escape from the crime scene and not.*

Killing can raise p_A only inside $\mathsf{Corr} \cap \mathsf{Esc}$: outside Esc the shaved minutes change no outcome, and outside Corr (below) killing *lowers* p_A. The five grounds below — grouped by which term of $(1 - p_A)\,W > c$ they act on — show the inequality fails.

Conditional on Corr*, the gain* $(1 - p_A)\,W$ *is minimal* (p_A *is already* ≈ 1)*:*

1. a well-trained fractal is recognized in $\approx$1 minute, so brute-forcing the last point saves $\mathcal{A}$ at most that $\approx$1–2 minutes over coercing it — bounding the entire interval on which Esc, and hence any gain from killing, can act;
2. $\mathcal{A}$'s plan already budgets time-lock or re-derivation, subdue, and interface time, so $\Pr[\mathsf{Esc}]$ is small: the residual minute rarely flips the escape, and killing pays off only inside Esc;
3. the terminal K is deliberately cheap ((5) keeps memory-hardness in the orbit, not the terminal step), so $\mathcal{A}$ can *pre-stage* the 2^{32} brute as a "victim-stalls" contingency — recovering the shaved minutes *without* killing, and draining the gain to ≈ 0.

Absent Corr*, killing makes* p_A ***fall****, not rise:*

4. if any surrendered point is wrong, $\mathcal{V}$ is $\mathcal{A}$'s only route to a corrected one; eliminating $\mathcal{V}$ burns the coercion retry and collapses p_A toward 0. Since $\mathcal{A}$ cannot verify Corr before broadcasting, the *expected* change in p_A from killing is dragged below its Corr-only value — and can turn negative.

And the cost c is large:

5. the killing sharply *worsens judicial liability* — murder aggravated by frivolous motive *and* by the victim's defenselessness, atop armed robbery — so c sits well above the minute-scale $(1 - p_A)\,W$ the first four grounds leave on the table.

Whether the window opens after the $(r_N{-}1)$-th or $(r_N{-}2)$-th point rests on the softest number of Stipulation 1 — inside the tolerated reasoning either way.

Net. The relevant quantity is *not* $(1{-}p_A)\,W$, which is non-negative by definition ($p_A \in [0, 1]$), but the *utility increment* of the assassinate-then-brute strategy over keeping the victim cooperating. Writing Δp_A for the change in $\mathcal{A}$'s win probability that strategy induces, it is rational only if $\Delta p_A\,W > c$, and

$$\Delta p_A \;=\; \underbrace{\Pr[\mathsf{Corr} \cap \mathsf{Esc}]}_{\text{gain (grounds 1–3): minimal}} \;-\; \underbrace{\Pr[\neg\mathsf{Corr}] \cdot q}_{\text{loss (ground 4): burned retry}} ,$$

where $q \in [0, 1]$ is the win probability a still-cooperating victim would restore — via a corrected point — and that killing forfeits. The gain term is small (Esc is unlikely, is realized only within Corr, and even then is bounded to a minute the brute can be pre-staged around); the loss term is a strict subtraction growing with $\Pr[\neg\mathsf{Corr}]$; and c is large (ground 5). Hence the criterion the strategy must satisfy, $\Delta p_A\,W - c > 0$, is false with overwhelming probability, so the kill strategy is — much more likely than not — *not* rationally advantageous.

In plain English, the residual deadly-race window lives only in the intersection of a wrench attacker who:

1. is **technically sophisticated** — able to mount the 2^{32} brute and drive the interface. *(Extrinsic to TGPO: granted to $\mathcal{A}$ by Kerckhoffs (Assumption 1), which already hands over the full protocol.)*
2. **landed the attack inside the o_N-live window** — timed precisely enough to catch stage N available, or resourced enough to sustain coercion until o_N was made available;
3. **subdued $\mathcal{V}$ before any reset** of the derivation or the TLP;
4. **judged, right after the second-to-last point, that ceasing the attack immediately was critical to escape** (Esc) — despite already having budgeted for a longer encounter (item 2);
5. **could rule out every already-coerced point being incorrect** (Corr); and
6. **accepted the liability of a doubly-aggravated murder** — frivolous motive *and* victim defenselessness. *(Once again extrinsic to TGPO: it burdens the deadly-racer attacker of any coercion-resistance protocol.)*

Remark 24 (Cloaking the window — the Jade Clock). The gain term carries a further hidden factor: the residual is reached only when the attack *coincides* with the brief interval in which o_N is live. That coincidence is itself suppressible. Anonymizing *when* the vault-opening window occurs — the role of the outsourced anonymous time-lock ("Jade Clock") [14] — drives $\Pr[\text{attack lands in the target } o_N\text{-live window}]$ down, and this factor multiplies the already-minimal $\Pr[\mathsf{Corr}{\cap}\mathsf{Esc}]$. Here anonymity of the *timing* suppresses a residual that confidentiality of the *secret* alone does not reach.

Remark 25 (Design decision — canonical-island highlighting). Item 5 above (that $\mathcal{A}$ cannot rule out $\neg\mathsf{Corr}$) can be strengthened at the UX layer, to make an *intentionally* wrong prior point plausible. Whenever feasible the interface should keep the canonical-island highlight active: this lets a coerced $\mathcal{V}$ perform a random zoom and then feign a confident, plausible-looking selection of an *incorrect* island. Without the highlight the victim could not reliably produce a plausible selection at all — valid coordinates are a tiny fraction of the space, so an unaided pick would most likely be rejected as invalid — so it is the highlight that makes a deliberately-wrong-yet-credible surrender available to the victim, and thus keeps $\neg\mathsf{Corr}$ live in $\mathcal{A}$'s belief (inflating the loss term of Δp_A). A sophisticated $\mathcal{A}$ could force a non-highlighting fork of the app, but that costs further sophistication and wrench-time. This layer sits *on top of* a window of deadly-race vulnerability that is already vanishingly unlikely (the six-way intersection above).

Remark 26 (Feigning is not load-bearing). The $\neg\mathsf{Corr}$ loss term (ground 4) and its UX amplification (canonical- island highlighting, above) only *deepen* the case; they are not what carries it. Grant $\mathcal{A}$ the best of it — $\Pr[\mathsf{Corr}] = 1$, an honest $\mathcal{V}$ in flawless cooperation (D3), with no feigning to fear. Then the loss term vanishes and the kill increment collapses to its gain term alone,

$$\Delta p_A \, W - c \;=\; \Pr[\mathsf{Esc}] \, W - c,$$

which grounds 1–3 already hold small ($\Pr[\mathsf{Esc}]$ tiny, the shaved interval minute-scale and pre-stageable) against a large c (ground 5). So the residual is bounded on the *gain* side by itself: feigning ($\neg\mathsf{Corr}$) subtracts further and is a welcome bonus, but the window is already small without it — even against a perfectly honest, fully cooperating victim.

Remark 27 (Splitting a stash across several K_i). Nothing forces a holder to route an entire stash through a single master secret; it may be split across several K_i — distinct stages, or independent orbit setups — with two risk-management benefits, each diluting one of the residual failure modes:

1. *Diluted seizure risk* — the proverbial "not all eggs in one basket." Should the residual *seizure* window (§4.6), or any other failure mode of the construction, expose one K_i, only the fraction of the stash it guards is lost.
2. *Diluted deadly-race payoff.* In the residual *deadly-race* window (§4.7) the kill condition is $(1 - p_A)\, W > c$; with the stash split, the W at stake in any single window is only a fraction of the whole, lowering $\mathcal{A}$'s assassination payoff and making the inequality harder to meet.

4.8 Empirical grounding in the registry

The two stipulations that lean on the physical-attack registry [2] — the coercion-time budget t (§8) and the murder-cost weighing of this section — admit directional empirical support. The registry records each incident only as *date, victim, location, headline*; it carries no duration or outcome field, so we code the 342 incident headlines (accessed 2026-07-19) by keyword. The coding is coarse and the categories overlap; a reproducible script accompanies the paper.[2] Three things stand out.

The attack is frequent and worsening. Recorded incidents climb from a handful a year before 2017 to a peak of 82 in 2025 — the growth the introduction cites, now counted rather than asserted.

The terminal move is real. At least 5.3% (18/342) of headlines record a *killed* or *murdered* victim. A homicide is usually reported as a homicide, not as a "Bitcoin attack", so this is a *lower* bound — yet even so it shows the assassination outcome the Deadly-Race Lemma (Lemma 1) predicts is attested in practice, and that the cost c is weighed against an action attackers demonstrably take.

Short attacks dominate; the registry over-counts the dramatic. Most wrench attacks are *hand-over-now* events — a threat, an immediate transfer, minutes on scene; the long, cinematic kidnapping is the exception. A media-sourced registry inverts this, because a mutilation-kidnapping is news where a knifepoint mugging is not, so the kidnap share coded below *over*-states the long tail rather than establishing it. That TKBA is realistically *less* effective against a long-duration attack — where hours of pen-and-paper effort under coercion *might* walk a deep stage — is granted, but it is *speculative* pending the field study of §8. What is *not* speculative is that the fielded alternatives fail; the next case makes that concrete.

[2] `registry-analysis/` in the source tree gives the exact keyword patterns, per-year counts, and caveats.

Signal (headline keyword coding)	Incidents	Share
Killed / murdered victim *(a lower bound)*	18	5.3%
Kidnap / hostage / ransom (long-hold)	133	38.9%
Armed robbery at gunpoint (short-hold)	93	27.2%
Home invasion / break-in / raid	35	10.2%
Torture or physical violence	78	22.8%

Categories overlap (a kidnapping may also be violent or fatal) and do not sum to 100%; shares are of all 342 incidents, and the modality labels are a headline proxy, not a measured duration. The sample is media-reported and survivorship-biased — deterred, unreported, and silently-fatal cases are absent — so every figure is directional and the fatality share in particular is a floor.

A case in point — Balland. The January 2025 abduction of David Balland, a co-founder of the hardware-wallet maker Ledger [15], is instructive precisely because it is an *embarrassment to the self-custody creed.* A professional champion of self-custody was seized from his home and mutilated — a finger severed to force payment — and neither wealth nor prominence prevented it. He was freed by an elite *state* police unit, and the ransom — paid not by Balland but by an associate — was clawed back largely because part of it moved in a *centralized* stablecoin whose issuer could freeze it. A self-custody pioneer made whole by the state and by centralization: two mechanisms external to, and in tension with, the creed. Only the issuer freeze is independently verifiable; the remainder rests on the parties' accounts, according to which roughly 5% was never recovered. The case also indicts the delegated route (R2): an associate *did* pay, which is exactly the compliance clause D8 assumes away — a real instance of the indirect coercion, routed through the victim's body to a third party who will not refuse. For a brief hand-over-now robbery a remote delegate is unreachable in time and so seems to help; but merely *not carrying* the device — leaving the wallet in a home vault — does the same, while for the long attack the delegate is not out of reach but is the ransom channel itself. The lesson is not that Balland erred: it is that, against a determined long-hold attacker, the fielded options collapse to state rescue and centralized freezing, whereas the tacit route removes the leverage — no standing device, no hostage token, no delegate to pay — rather than relying on ex-post recovery.

5 Classification of existing designs

Corollary 2 gives a classification axis: a design that clears Criterion 1 does so either by *removing the race* or by *removing $\mathcal{A}$'s reach over the racer.* The framework should classify shipping designs, and it does — via the reachability corollary, not by calling them broken.

5.1 Time-lock-rescue vaults

Take **Rewind Bitcoin** [16] as the exemplar. Funds sit in a Bitcoin-native time-lock vault; unlocking starts a multi-day countdown; if an unauthorized unlock is triggered, the owner *or a trusted delegate* has that window to press "Rescue" and sweep to a cold panic address.

Against *remote* theft it is sound: a thief who steals keys trips the countdown, the owner notices, and rescues. But the spend capability sits behind *only a delay*, with keys on the seized device and no recall floor — so a wrench attacker who seizes the device *does* hold a feasible path (unlock, wait out the countdown). That is a race, in the sense of Lemma 1. Rewind's answer is exactly the second route of Corollary 2: it puts the winning racer out of $\mathcal{A}$'s reach — a *remote delegate* the attacker cannot find or eliminate presses Rescue — so the attack fails cleanly. This is a legitimate clearance, but the remote-racer kind: its cost is that **custody is no longer individual**.

Remark 28 (When the escape closes). The remedy holds only while the delegate is genuinely remote. If the delegate is a friend who happens to be visiting during the attack, *both* problems apply at once — custody was already non-individual, and the racer is now co-located and reachable — so the design becomes deadly-race-susceptible again. The user is thus forced into an awkward corner: trust someone they can rarely be near, and accept a perverse incentive to avoid physical proximity to a trusted friend. The same closure afflicts the remote-delegate route to a perceptual oracle (Remark 17).

5.2 A second time-lock design — BitVault

BitVault [17] is the same species with a different rescue verb. Funds sit in a multisig with *time-delayed transactions* (a configurable 2 h–15 day window); a spend enters the delay, an alert reaches the owner's chosen "owl wallet," and the owner may *cancel* before settlement. It is marketed as coercion-resistant self-custody — it "protects your Bitcoin even if you're physically forced to hand over your keys."

The structure is Rewind's, with cancel-during-delay standing in for Rescue: so against remote theft it is equally sound, and against a wrench attack it is equally a race (Lemma 1) — the spend sits behind only a delay, keys on the seized device. The difference is *who the racer is*. Rewind names a *remote delegate*; BitVault routes cancellation to the *owner's own* alert wallet. Under coercion the owner is captive and, by D3–D4 (Assumption 2), that wallet is seized or coerced with everything else — so by default the racer is *co-located and reachable*, and the design is deadly-race-susceptible. It clears Criterion 1 only if the owl wallet is a genuinely remote, autonomous rescuer (the honest R2, shared custody); yet it advertises **individual** self-custody ("zero counterparty risk," "non-custodial"). That is the one combination the reachability corollary forbids — a recall floor (R1, individual) *or* a remote racer (R2, shared), never individual custody with a reachable racer.

Two auxiliary claims do not close the gap. First, that time-delayed transactions are "publicly verifiable, making kidnapping attempts pointless" is a *deterrence* claim — about whether the attack occurs — not a structural one about the game once it does; under Kerckhoffs a WDY attacker who *knows* the delay plans around it: coerce the cancel key up front, outwait a 2 h window, or — since the canceler is the competing racer — eliminate them to stop the cancellation, the very incentive of Lemma 1. "Cannot be moved *instantly*" is not "cannot be moved," and advertising pointlessness is the false comfort the Lemma's remarks name: danger peaks where the marketing promises safety. Second, the "decoy wallet" is a plausible-deniability layer, which fails Kerckhoffs (Assumption 1) against an educated attacker who persists past the decoy. *(This reading is drawn from BitVault's public description; if its cancel path is in fact a pre-signed sweep triggered by a genuinely remote, un-coercible watchtower, it is simply the honest R2 case at shared-custody cost, and the critique narrows to the individual-custody configuration it markets.)*

A further caveat sharpens R2 itself: it needs the winning racer to be not merely remote but genuinely *un-coercible* — an autonomous watchtower, not a human friend. Where the delegate is a person, coercion simply routes through the victim to them, and a humane delegate pays; the Balland case (§4.8) is exactly this failure, and it is the reality clause D8 idealizes away.

5.3 TGPO, and the honest comparison

TGPO clears by the *other* route: the secret is tacit and never on the device (Assumption 3, Theorem 3), so a seizure yields nothing feasible — there is no racer to reach, and the time-lock is only a delay *on top of* an already-clean failure. Read through the Four Properties (Corollary 3), the comparison is a covering-set statement: each shipping design buys (or tries to buy) coercion resistance by *trading at least one* property TGPO keeps — and one of them trades a property and *still* fails. TGPO is the only design holding all four.

Design	(1) Know.	(2) Indiv.	(3) Non-ob.	(4) Coerc.	What it trades
TGPO (Great Wallet)	✓	✓	✓	✓	holds all four (tacit; R1)
Rewind Bitcoin	×	×	✓	✓	individual custody: a remote delegate (R2)
BitVault	×	✓	×	×	non-obscurity: assumes attacker ignorance (± geography) — gray, can backfire
Casa	✓	×	✓	✓	custody shared with remote co-signer

Row (1) is the covering point: TGPO is the *only* knowledge-based (deviceless — the secret is knowledge, never a device or a location) row, because every competitor keeps a *transmissible* secret on a device. **Rewind** trades individual custody for a remote delegate (R2). **Casa**'s collaborative multisig — the client's keys on separate hardware plus a Casa co-signing key — trades individual custody: the co-signer is a remote party, and a coercion-resistant delegate is remote *by definition*, so the cost is individual custody, not non-geography (Casa asks the user to visit no site of its own) [18]. **BitVault** trades non-obscurity (its cancel and decoy assume attacker ignorance, failing Kerckhoffs) and, per §5 above, still lands in the gray zone — the red × in column (4) is *negative* coercion resistance (it can backfire), not mere absence. The axis tilts further to R1 for a blunter reason (Remark 9): the delegated rows are not self-custody at all — their clearance assumes a delegate that refuses even the legitimate owner — and, the rescue capability being *transmissible*, they keep a clean-loss exposure the tacit route does not. *Not your keys, not your coins.*

6 Quantum note

Memory- and depth-hardness is a tunable lever, and even its modest settings suffice here. A quantum search over an n-bit space needs $\sim 2^{n/2}$ *coherent* evaluations of the oracle [19], and each such evaluation is a deeply sequential Argon2d computation whose full *depth* the coherence must survive — a cost the quantum resource-estimation literature already finds decisive even against oracles far cheaper than ours [20]. Even a *few-minutes* H^* per element therefore places Grover out of reach; an *hours-scale* H^* does so *a fortiori*. Hence sub-256-bit standards (96 or 128 bits per stage) are quantum-defensible *without* a 256-bit seed — the memory- and depth-hardness behind the floor's *hard* bound (64^*) classically also caps the quantum speed-up.

Remark 29 (The one bound Grover does dent: the live-o_N residual)*.* The duality's *cheap* bound is a different matter. The floor also clears 64^* via a cheap 2^{96} ledger brute of Sh_N (Stipulation 1, S3), and that cheap side inherits *no* depth-hardness to guard it: Grover halves it outright to $\sim 2^{48}$. Concretely, a Quantum WDY attacker landing on an imminently- or currently-live final board o_N — the same coincidence that opens the tolerated residual window (§4.6) — faces only the cheap 2^{96} brute of the terminal secret, and Grover reduces that to a $\sim 2^{48}$ ("48-qubit") floor.[3] This touches nothing but the residual: it does *not* reach the memory-hard 64^* bound (guarded above), and it applies only inside the already-ε-bounded live-o_N sliver, whose incentive-safety argument (§4.7) turns on the kill economics, not on the brute's width — a 2^{48} Grover search buys $\mathcal{A}$ nothing a classical 2^{32} brute in that same window does not already grant.

7 Related work

[3]Each Grover attempt is one cheap evaluation of the terminal step — a single $\mathsf{H}(o_N \,\|\, Sh_N)$ key-derivation followed by the ledger decryption/spend check — and *nothing* more: the search is over the 96-bit Sh_N *directly*, so Shamir reconstruction (the holder's step of interpolating Sh_N from r_N recognized points) never enters the oracle loop. One quantum attempt is thus KDF + decryption, not Shamir + KDF + decryption.

Coercion resistance. The term is due to Juels, Catalano, and Jakobsson [21], in electronic voting: a scheme is coercion-resistant if a voter *cannot prove* to a coercer how they voted, so the coercer cannot verify compliance and the reason to coerce collapses. Our sense is different and complementary. There the protected act is a *choice* the coercer wants to dictate and safety comes from *unprovability*; here the protected object is a *bearer secret* the coercer wants to extract, and the danger is not that compliance is unverifiable but that a released holder remains a racer worth killing (Lemma 1). We inherit the word, not the model.

Secrets recognized but not articulated. The tacit premise (Assumption 3) — a secret its holder can *recognize* but cannot *transmit* — has direct antecedents. Bojinov, Sanchez, Reber, Boneh, and Lincoln [22] train an authentication secret by *implicit learning* so that even a cooperating, coerced user cannot articulate it ("rubber-hose" resistance); this is the closest prior instantiation of our TKBA, and TGPO can be read as carrying the same idea from an authentication token to a self-custody key behind a memory-hard time-gate. The broader phenomenon — identity carried in a behavior that resists articulation — is old: keystroke dynamics treat idiosyncratic typing rhythm as a biometric [23], and WWII traffic analysis identified individual wireless operators by the "fist" of their Morse keying long before either was formalized [24].

Deniability versus unprovability. Deniable encryption [25] and the duress-PIN / panic-password schemes it inspired let a coerced user *plausibly lie* — surrender a fake key, or open a decoy. NNPP (Assumption 4) is the sharp negative counterpart: rather than proving a false positive ("this is the key"), the holder would need to prove a true negative ("no other key exists"), which cannot be done. Deniability offers a lie the coercer cannot disprove; we show the holder cannot furnish the assurance the coercer would need, which is exactly why the coercer's only remaining lever is the victim (§3). Note too (§3.6) that decoy and duress schemes rest on the adversary's ignorance and so fail Kerckhoffs.

Time-gates. The construction's wall-clock gate is a time-lock puzzle in the sense of Rivest, Shamir, and Wagner [13]; verifiable delay functions [26] are the modern, publicly checkable form of the same primitive, and either may instantiate the outsourced time-lock the orbit optionally uses (§4).

8 Limitations and scope

The tacit premise is unproven — the softest assumption. Everything on the tacit route (hence "TGPO keeps all Four Properties") rests on the existence of a *perceptual oracle*: a stimulus a human can *recognize* but not *articulate*. For TKBA it must simultaneously (i) admit a *binary encoding*; (ii) be *feasible to memorize* at the stipulated minimum ($\geq$ 96 bits); and (iii) be *infeasible to transmit within the coercion-time budget t* (Definition 12). None of the three is proved here; their conjunction is an empirical, psychometric claim awaiting a field study. Perturbed Burning-Ship fractal locations are our current instantiation — the best present candidate, with the design record giving the grounds for expecting them to hold — but until a study establishes such an oracle, **the validity of TKBA remains an assumption, not a result**, and it is the softest premise in the paper. Its imperfection is precisely the residual window (§4.6), where transmission becomes possible near the end of the derivation.

Memorization cost, and why it is affordable. A separate objection to a brain-only setup is not whether the oracle *exists* but whether living with it is too costly — carrying $\geq$ 96 bits of tacit recall indefinitely. Two things bound that cost. First, retention is *assisted*: the *Celestial Peace — Never Forget* (CPNF) companion [27] schedules spaced-repetition review (an SM-2 / Anki-class algorithm), turning a one-off feat of memory into a light, decaying maintenance load. Second, the feat is historically ordinary: before mobile computation offloaded it, ordinary people routinely held a handful of phone numbers, several street addresses, and a working mental map of their city's grid — recall comparable to what TKBA asks. The cost is real but bounded, and it is the price of a secret no device can be seized to reveal.

Stipulations aligned with standard practice. Two quantities are stipulated, not derived. (a) The floor rests on the feasibility line S1, S2, S3 (Stipulation 1) — a standard *key-size* stipulation (2^{64} memory-hard and 2^{96} cheap evaluations are beyond reach, in line with accepted security-level conventions); we do *not* claim a reduction to a named hard problem, and none is needed for a key-sizing argument. (b) The coercion-time budget t is a stipulation from *kidnapping-duration statistics* — read off the physical-attack registry [2], with the common-sense monotonicity that a kidnapping's material feasibility *falls* with its duration. A *long-hold* adversary who sustains coercion past the time-gate ($t > G$) is therefore out of scope: the tacit route assumes $t < G$, and against unbounded patience no in-head time-gate suffices. Most recorded attacks are in fact short hand-over-now events (§4.8).

Single-shot utility, and why it is the conservative case. The kill decision is modeled once, risk-neutrally. Risk-neutrality is *worst-for-victim*: a risk-averse $\mathcal{A}$ discounts the uncertain kill-gain $(1 - p_A)W$ and so kills *less* readily, while the cost side is already carried by c — so the single-shot risk-neutral model is a fortiori safe. *Repeated* encounters are absorbed by the construction's re-key pathway: a victim who survives with K uncompromised — itself a spectacular outcome, achieved with no delegate and no geographic split — has every incentive to harden physically for the short time needed to *re-key* the stash to a fresh setup (the level-up / shallow-stage-retirement pathway). *Inheritance* we leave beyond scope except for its design target: the protocol should render *assassinate-then-coerce-heir* non-viable — simplest remedy, a dead-man switch with heirs *also* TGPO-protected, so coercing an heir faces the same floor.

The one-line takeaway. Under TGPO — or a geographic setup, available only to those who can hold multiple independent sites — you *survive a WDY attack with individually custodied K intact.* Alternatively you resort to shared custody, in which case, *unless* your stash is *demonstrably* more available to some remote (trusted) delegate than to you, the outcome of a wrench attack is "gray" — hence, by the No-Gray-Area Criterion, a sufficiently vicious attacker, or a high enough stash ($W > c/\pi$), gets you killed.

Composition across level-up. DRL-safety across the setup pathway (re-keys, retirement-via-chain, a second wrench attacker who effectively starts a stage deeper) is a *deduction* from the per-encounter results applied at each stage, not a fresh theorem; we note it without re-proving it, and leave a full multi-encounter treatment to the utility-model extension above.

Also out of scope. Adversarial-agent scenarios (testator/heir mutual distrust, malicious interference with rotation); device side-channels (timing, EM, acoustic) during active tacit recall; long-horizon cryptographic erosion; and mass on-chain correlation beyond standard taproot / private-channel privacy. Finally, the framework fixes a *bar* (Criterion 1); it does not claim TGPO is the unique construction that clears it, and the residual window (§4.7) is a genuine bounded sliver argued incentive-safe rather than eliminated.

9 Ethics and responsible disclosure

This is offensive-flavored security research with direct life-safety stakes: we formally derive an *assassination* incentive (Lemma 1). We therefore state plainly why publishing it is, on balance, protective.

The incentive is disclosed, not created. The Deadly Race is a property of a situation a coercer already faces; naming it hands nothing to an attacker who, by Kerckhoffs (Assumption 1), is already assumed to know the protocol and the landscape. What publication changes is only that *defenders* can now measure a design against the bar (Criterion 1) and reject the ones that manufacture the incentive. We describe an incentive and the failure modes of existing defenses — not a targeting method, and no operational how-to; the paper's constructive content is defensive.

Obscurity is not a sound reason to withhold. The alternative — not publishing analyses of attack vectors and the failure modes of deployed defenses — is itself a bet on obscurity, and both theory and data say that bet loses. As Kerckhoffs' principle has held for a century, a defense that works only while the adversary stays ignorant provides something worse than defenselessness: *a false sense of security.* Empirically, the physical-attack registry [2] and the accompanying economic analysis [5] show exactly what the principle predicts — wrench attacks are frequent and rising, and the community's entrenched counsel (*keep a low profile*, *never disclose ownership*, *deny if asked* — advice prominent channels actively promote, catalogued in [5]) has not stemmed them. Security knowledge diffuses to criminal actors regardless of academic reticence; withholding it handicaps only the defenders who would otherwise design against the very incentive we name. Publishing the Deadly Race, *together with* a bar that rejects the designs which create it, is net-protective of the holders it concerns.

Framing and restraint. We keep the framing on protecting at-risk holders, disclose our own bounded residual honestly (§4.6) rather than overclaim, and treat the shipping alternatives we classify (§5) as legitimate clearances by the other route — not targets.

10 Conclusion

The Deadly Race turns "graceful degradation" from a virtue into a liability whenever the degraded state is a feasible-but-slower path to the secret: by Lemma 1, that state is a race, and the race pays an assassination incentive. The No-Gray-Area Criterion is the resulting design bar, and the reachability corollary shows there are exactly two honest ways to clear it. The TGPO orbit clears it by removing the race entirely — a floor that leaves K prohibitive to reach from every seizable state bar a bounded residual (Theorem 3, §4.6) — and so keeps custody individual, whereas time-lock-rescue vaults clear the same bar by removing the attacker's reach at the cost of a trusted third party.

What devicelessness buys: a threat model that stays simple. Beyond clearing the bar, keeping the whole setup in the head keeps a series of otherwise-open cans of worms firmly shut — and it is this that lets the model above stay *a fortiori* and single-shot rather than a tangle of sub-games. The tacit, deviceless TGPO holder enjoys:

1. a **minimal window** in which the stash K is seizable at all — the bounded residual (§4.6), not a standing device *(the funds survive)*;
2. a *separate* **minimal window** of deadly-race exposure *(the owner survives)* — distinct from (1), and only mostly overlapping it: once the second-to-last point of stage N is surrendered, the stash is effectively lost while the deadly-race window is at its *peak* (an impatient $\mathcal{A}$, or one perceiving a threat to its escape, may still kill);
3. **no hostage device** — and so none of the game-theoretic bluffing maze of ransoming a seized token (§3.7);
4. an NNPP that is a **dead end**: $\mathcal{A}$ may assume an undisclosed shortcut device exists (Assumption 4), but killing $\mathcal{V}$ cannot extract it, and $\mathcal{A}$ knows $\mathcal{V}$ has no *need* of one — so denial is inherently plausible (Remark 22);
5. **no trusted third party** (route R1, not R2);
6. **maximum portability** — no geographic binding to sites of custody (Definition 11);
7. the **lowest operational cost** of the three routes.

Each is a simplification that device-, geography-, or delegate-dependent designs forfeit; taken together they are why the deviceless secret admits so plain a threat model.

Open items for the full draft. The impossibility chain is drafted here (§3); the remaining work is: an *explicit adversary game* — $\mathcal{A}$ a PPT adversary augmented by a bounded physical-coercion oracle — making Criterion 1 a theorem rather than a stipulated bar; a utility model richer than the single-shot $(1-p_A)\,W > c$ (repeated encounters, partial extraction, heir cascades,

$\mathcal{A}$'s risk-aversion); the floor *reduction* discharging §4's $p_A = 0$ obligation; a quantitative H^* per-evaluation calibration tying S1/S2 to hardware; and an empirical grounding of the residual-window argument against the wrench-attack registry [2].

A A game-based formulation

The body states coercion-safety in a *decision-theoretic* register: the coercion game $\mathbf{Coerce}_\Sigma$ (Definition 5) asks whether a *priced* adversary WDY$[W, c]$ ever strictly prefers to kill. Readers from the cryptographic tradition will expect the complementary *experiment-and-advantage* form. We give it here — and, mirroring the two axes the residual splits into (§4.6 for funds, §4.7 for the deadly race), we give *two* experiments over one shared WDY interaction: a **seizure game**, whose winning event is the familiar "the adversary reaches K," and a **deadly-race game**, whose winning event is the novel "the adversary reaches a *gray* state" — the one that manufactures the kill incentive (Lemma 1), and which an adversary can trigger even when it provably cannot learn K.

Two adversaries, and why the primed one is the worst case. The seizure game keeps the body's adversary $\mathcal{A}$ (holder $\mathcal{V}$): it wants the *money*, seeks K, and would kill only *instrumentally*, where the priced calculus $(1 - p_A)\, W > c$ makes a kill raise its take (Definition 4). Its question is fund-safety — *does $\mathcal{A}$ reach K?* The deadly-race game instead posits a *hypothetical* adversary $\mathcal{A}'$ (holder $\mathcal{V}'$, primed to keep the two games apart) that wants the deadly race *for its own sake*: it plays to reach a gray state. This is deliberately the worst case for the kill question. Reaching a gray state is *necessary* for any kill to pay (Lemma 1); so if even the DR-hunting $\mathcal{A}'$ cannot reach one — the coercion-resistance result below, $\Pr[\text{win}] = 0$ — then the money-seeking $\mathcal{A}$, who would kill only instrumentally, never lands in a state where killing $\mathcal{V}$ pays. *$\mathcal{A}'$ trying and failing to manufacture a deadly race is exactly what certifies that $\mathcal{A}$ will not run one.* Throughout, p_A and π stay unprimed: they are the body's state functionals (Definitions 2, 3), read off the resulting position, and each game is a different *condition* on them.

A.1 The shared interaction

Definition 13 (The WDY interaction)**.** Fix a custody protocol Π, a WDY adversary $\mathcal{X}$ (Assumption 2), and the honest holder $\mathcal{Y}$.

1. **Setup.** The challenger runs Π's honest setup for $\mathcal{Y}$, producing the master secret K, the co-located device and explicit-secret state D (vault and time-lock blobs, commitment scripts, review schedules), and the tacit state T (the holder's recognition capability). Public parameters pp, including the salt σ, are given to $\mathcal{X}$ (clause D0).
2. **Interaction.** Within a single attack site, $\mathcal{X}$ queries, in any order:
   - Seize — returns the co-located device and explicit-secret state D (clauses D4,D5).
   - $\mathsf{Coerce}(t)$ — for a coercion budget t, returns every value $\mathcal{Y}$ is *able* to transmit under full cooperation (clause D3); the tacit state T is available only as an *evaluate-only* oracle the challenger runs on $\mathcal{Y}$'s behalf, never as a transmissible value (clauses D6,D7).
   - $\mathsf{Compute}$ — any computation on the feasibility line (clause D1): at most 2^{64} cheap and strictly fewer than 2^{64} memory-hard evaluations.

   Any remote delegate is not coercible through $\mathcal{Y}$ (clause D8).
3. **Resulting position.** Let R be the resulting set of racers, $p_A = p_A(R)$ the probability $\mathcal{X}$ then reaches K (Definition 5), and $\pi = p_A(R \setminus \{\mathcal{Y}\}) - p_A(R)$ the holder's pivotality. Each game below is a predicate on this (p_A, π).

A.2 The seizure game — fund-safety

Definition 14 (Seizure experiment; seizure-resistance)**.** The **seizure experiment** $\mathsf{Sz\text{-}Exp}_\Pi(\mathcal{A})$ runs the interaction (Definition 13) with adversary $\mathcal{X} = \mathcal{A}$ and holder $\mathcal{Y} = \mathcal{V}$, and outputs 1 ("$\mathcal{A}$ *seizes*") iff $\mathcal{A}$ feasibly reaches K — equivalently $p_A > 0$ (Definition 2). Π is **seizure-resistant** if $\Pr[\mathsf{Sz\text{-}Exp}_\Pi(\mathcal{A}) = 1] = 0$ for every WDY adversary $\mathcal{A}$.

This is the fund-safety / seizability axis — the game face of the *floor*.

Corollary 4 (TGPO is seizure-resistant up to the residual)**.** *The Time-Gated Perceptual Oracle (§4) is seizure-resistant except on the residual* seizure *window (§4.6).*

Proof. By Theorem 3, from every seizable state save that window K_N meets the floor, so no feasible path to K exists and $p_A = 0$; hence $\mathsf{Sz\text{-}Exp}$ outputs 0. On the residual seizure window $\mathsf{Sz\text{-}Exp}$ may output 1 — the o_N-live coincidence where $\mathcal{A}$ does reach K; the loss there is of *funds*, bounded as argued in §4.6. □

A.3 The deadly-race game — deadly-race-safety

Definition 15 (Deadly-race experiment)**.** The **deadly-race experiment** $\mathsf{CR\text{-}Exp}_\Pi(\mathcal{A}')$ runs the interaction (Definition 13) with adversary $\mathcal{X} = \mathcal{A}'$ and holder $\mathcal{Y} = \mathcal{V}'$, and outputs 1 ("$\mathcal{A}'$ *wins*") iff $\pi > 0$; for the individual-custody case $R = \{\mathcal{V}'\}$ this is exactly a *gray* outcome, $0 < p_A < 1$ (Definition 3).

Definition 16 (Coercion resistance)**.** Π is **coercion-resistant** if $\Pr[\mathsf{CR\text{-}Exp}_\Pi(\mathcal{A}') = 1] = 0$ for every WDY adversary $\mathcal{A}'$.

Remark 30 (Where the two games part — the clean loss)*.* The two experiments read the same position (p_A, π) but score it differently, and they diverge at exactly one point. At $p_A = 0$ both output 0 (no feasible path: the defender clears both). On the *gray* interval $0 < p_A < 1$ both output 1 ($\mathcal{A}$ holds a path and $\mathcal{V}'$ is pivotal: the defender fails both). They part only at *full compromise* $p_A = 1$: the seizure game outputs 1 (K is gone) while the deadly-race game outputs 0 ($\pi = 0$ — with the adversary already certain to win, removing the holder changes nothing, so no kill incentive arises). That single disagreement is the whole content of *"a clean loss is deadly-race-safe"* (Remark 9): losing the funds outright is a seizure-game defeat but a deadly-race-game *non*-event. Conversely — the paper's negative result restated — an adversary that provably *cannot* win the seizure game (cannot reach K; e.g. one facing a graceful-recovery vault it cannot open within the custody window) can still win the *deadly-race* game by reaching a gray state. The deadly-race axis is therefore *not* implied by fund-safety; it is a separate axis, exactly as §4.6 and §4.7 are separate. Fund-safety the construction handles on its own terms — *on top of* removing the gray race, it minimizes the window in which the stash is seizable at all, via the wipe-before-memory-hard step of (4) and the floor (Theorem 3).

Remark 31 (The residual window: from 0 to ε)*.* Definition 16 is the information-theoretic, no-gray-area form. The residual window (§4.7) replaces the exact 0 by a bounded $\varepsilon = \Pr[\mathsf{Corr} \cap \mathsf{Esc}]$: call Π **ε-coercion-resistant** if $\Pr[\mathsf{CR\text{-}Exp}_\Pi(\mathcal{A}') = 1] \le \varepsilon$ for all $\mathcal{A}'$. The claims below are the $\varepsilon = 0$ statements; the residual makes them ε-small — not zero — in the sliver argued incentive-safe there. In plain terms, this sliver is the moment the attack coincides with a *live* final orbit point o_N — currently or imminently materialized — and $\mathcal{A}'$ prevents the wipe (or reset) that would otherwise close it; only then is transmissible material briefly within reach, which is why the window is a bounded ε rather than a standing exposure.

A.4 Consistency with the main results

Theorem 4 (The experiment coincides with the Criterion)**.** Π *is coercion-resistant (Definition 16) if and only if* $\pi = 0$ *at every reachable state — equivalently, iff* Π *satisfies the No-Gray-Area Criterion (Criterion 1), iff* Π *is coercion-safe in the decision-theoretic game (Definition 5).*

Proof. CR-Exp outputs 1 exactly on a reachable state with $\pi > 0$, so $\Pr[\text{win}] = 0$ iff no reachable state is pivotal. By Theorem 1, $\pi = 0$ everywhere is equivalent to coercion-safety and to Criterion 1. The two clearance routes of Corollary 2 — individual custody (R1) and a remote un-coercible racer (R2) — are exactly the two ways to force $\Pr[\text{win}] = 0$. □

Corollary 5 (TGPO is coercion-resistant)**.** *The Time-Gated Perceptual Oracle (§4) is coercion-resistant.*

Proof. The experiment outputs 1 only on a *gray* state ($\pi > 0$); both clear endpoints $p_A \in \{0, 1\}$ give $\pi = 0$ and hence output 0. By Theorem 3 every state reachable by Seize, Coerce, and Compute leaves K at the floor, so the reachable endpoint is the *clear failure* $p_A = 0$ — the opposite endpoint $p_A = 1$ would require feasibly reaching K, which the same theorem bars. Either way $\pi = 0$, so the experiment never outputs 1 and $\Pr[\text{win}] = 0$. The only place p_A rises above 0 (to 1 in the worst instant) is the residual window (§4.6), where the tacit premise momentarily fails; accounting for it yields ε-coercion resistance. □

Remark 32 (Reading the classification as adversaries)*.* Under Definition 16, the classification of §5 is a statement about which designs admit a winning $\mathcal{A}'$. Against a time-lock-rescue vault advertised as *individual* custody, the adversary that calls Seize and then waits out the delay is winning: the rescue racer is co-located and reachable (clause D4), so $0 < p_A < 1$ and $\pi > 0$. Routing rescue to a genuinely remote, un-coercible delegate removes the reachable racer and restores $\Pr[\text{win}] = 0$ — the honest R2 case, at the cost of individual custody.

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